

LMR36510 SIMPLE SWITCHER® 4.2V to 65V, 1A Synchronous Step-Down Converter

1 Features

- Designed for reliable and rugged applications
 - Input transient protection up to 70V
 - Protection features: thermal shutdown, input undervoltage lockout, cycle-by-cycle current limit, hiccup short-circuit protection
- Well-designed for scalable industrial power supplies
 - Pin compatible with:
 - LMR36520 (65V, 2 A)
 - LMR33610, LMR33620, LMR33630, LMR33640 (36V, 1A, 2A, 3A, or 4A)
 - Internal compensation helps reduce design size, cost, and design complexity
 - 400kHz frequency
- Wide conversion range
 - Input voltage range: 4.2V to 65V
 - Output voltage range: 1V to 95% of V_{IN}
- Low-power dissipation across load spectrum
 - 90% efficiency: 24 V_{IN} , 5 V_{OUT} , 1A at 400kHz
 - Increased light-load efficiency in PFM mode
 - Low operating quiescent current of 26 μ A
- Power-good output with filter and delayed release

2 Applications

- IP network cameras
- Analog security cameras
- HVAC valve and actuator control
- AC drive and servo drive control modules
- Analog input modules and mixed IO modules
- General purpose wide V_{IN} power supply

3 Description

The LMR36510 regulator is an easy-to-use, synchronous, step-down DC/DC SIMPLE SWITCHER power converter. With integrated high-side and low-side power MOSFETs, output current up to 1A is delivered over a wide input voltage range of 4.2V to 65V. The transient tolerance that goes up to 70V reduces the design size and cost to protect against overvoltages and meets the surge immunity requirements of IEC 61000-4-5.

The LMR36510 uses peak-current mode control to provide excellent efficiency and output voltage accuracy. Precision enable gives flexibility by enabling a direct connection to the wide input voltage or precise control over device start-up and shutdown. The power-good flag, with built-in filtering and delay, offers a true indication of system status, eliminating the requirement for an external supervisor.

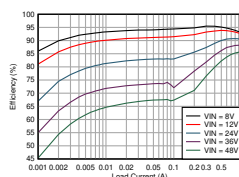
Integration and internal compensation eliminates many external components and provides a pinout designed for a simple PCB layout. The feature set of the device is designed to simplify implementation for a wide range of end equipment. The LMR36510 is pin-to-pin compatible with the LMR36520 (65V, 2A) and LMR33610, LMR33620, LMR33630 and LMR33640 (36V, 1A, 2A, 3A, 4A), completing the latest family of SIMPLE SWITCHER power converters. This compatibility increases the ease of use and scalability of wide V_{in} converters across various commonly used voltage and current ratings without having the need to redesign the board layout. As a result, the overall cost, design effort, and time to market are optimized.

The LMR36510 is in an 8-pin HSOIC package.

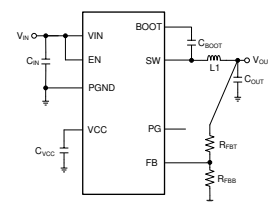
Package Information

PART NUMBER	PACKAGE ⁽¹⁾	PACKAGE SIZE ⁽²⁾
LMR36510	DDA (HSOIC, 8)	4.9mm × 6mm

- For more information, see [Section 11](#).
- The package size (length × width) is a nominal value and includes pins, where applicable.



Efficiency versus Output Current $V_{OUT} = 5V$, 400kHz



Simplified Schematic



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4 Device Comparison Table

ORDERABLE PART NUMBER	CURRENT	FPWM	f _{sw}
LMR36510ADDAR	1 A	No	400 kHz
LMR36510FADDAR	1 A	Yes	400 kHz

5 Pin Configuration and Functions

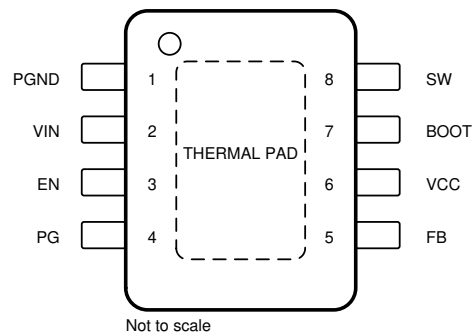


Figure 5-1. 8-Pin HSOIC DDA Package (Top View)

Table 5-1. Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO.		
PGND	1	G	Power and analog ground terminal. Connect to bypass capacitor with short, wide traces. Ground reference for internal references and logic. All electrical parameters are measured with respect to this pin.
VIN	2	P	Input supply to regulator. Connect a high-quality bypass capacitor or capacitors directly to this pin and PGND.
EN	3	A	Enable input to regulator. High = ON, low = OFF. Can be connected directly to VIN; <i>Do not</i> float.
PG	4	A	Open-drain power-good flag output. Connect to suitable voltage supply through a current limiting resistor. High = power OK, low = power bad. Flag pulls low when EN = Low. Can be left open when not used.
FB	5	A	Feedback input to regulator. Connect to tap point of feedback voltage divider. <i>Do not</i> float. <i>Do not</i> ground.
VCC	6	P	Internal 5-V LDO output. Used as supply to internal control circuits. Do not connect to external loads. Can be used as logic supply for power-good flag. Connect a high-quality 1-μF capacitor from this pin to PGND.
BOOT	7	P	Bootstrap supply voltage for internal high-side driver. Connect a high-quality 100-nF capacitor from this pin to the SW pin.
SW	8	P	Regulator switch node. Connect to a power inductor.
PAD	THERMAL PAD	Thermal	Major heat dissipation path of the device. A direct thermal connection to a ground plane is required. The PAD is not meant as an electrical interconnect. Electrical characteristics are not specified.

A = Analog, P = Power, G = Ground

6 Specifications

6.1 Absolute Maximum Ratings

Over junction temperature range of -40°C to 150°C (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
Input voltage	VIN to PGND	-0.3	70	V
Input voltage	EN to PGND	-0.3	70.3	V
Input voltage	FB to PGND	-0.3	5.5	V
Input voltage	PG to PGND	-0.3	20	V
Output voltage	SW to PGND	-0.3	70.3	V
Output voltage	SW to PGND less than 10-ns transients	-3.5	70	V
Output voltage	CBOOT to SW	-0.3	5.5	V
Output voltage	VCC to PGND	-0.3	5.5	V
Junction Temperature T _J		-40	150	°C
Storage temperature, T _{stg}		-65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

6.2 ESD Ratings

			VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM) ⁽¹⁾	±2500	V
		Charged-device model (CDM) ⁽²⁾	±750	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

Over the recommended operating junction temperature range of -40°C to 125°C (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
Input voltage	VIN to PGND	4.2	65	V
	EN to PGND ⁽²⁾	0	65	V
	PG to PGND ⁽²⁾	0	18	V
Output voltage	V _{OUT}	1	28	V
Output current	I _{OUT}	0	1	A

- (1) Recommended operating conditions indicate conditions for which the device is intended to be functional, but do not specify specific performance limits. For specified specifications, see *Electrical Characteristics*.
(2) The voltage on this pin must not exceed the voltage on the VIN pin by more than 0.3 V.

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		LMR36510	UNIT
		DDA (HSOIC)	
		8 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	42.9	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	54	°C/W
R _{θJB}	Junction-to-board thermal resistance	13.6	°C/W
ψ _{JT}	Junction-to-top characterization parameter	4.3	°C/W
ψ _{JB}	Junction-to-board characterization parameter	13.8	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	4.3	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application note.

6.5 Electrical Characteristics

Limits apply over operating junction temperature (T_J) range of –40°C to +125°C, unless otherwise stated. Minimum and Maximum limits⁽¹⁾ are specified through test, design or statistical correlation. Typical values represent the most likely parametric norm at T_J = 25°C, and are provided for reference purposes only. Unless otherwise stated, the following conditions apply: V_{IN} = 24 V.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
SUPPLY VOLTAGE (VIN PIN)						
I _{Q-nonSW}	Operating quiescent current (non-switching) ⁽²⁾	V _{EN} = 3.3 V (PFM variant only)		26	36	μA
I _{SD}	Shutdown quiescent current; measured at VIN pin	V _{EN} = 0 V		5.3		μA
ENABLE (EN PIN)						
V _{EN-VCC-H}	Enable input high level for V _{CC} output	V _{ENABLE} rising			1.14	V
V _{EN-VCC-L}	Enable input low level for V _{CC} output	V _{ENABLE} falling	0.3			V
V _{EN-VOU-H}	Enable input high level for V _{OUT}	V _{ENABLE} rising	1.157	1.231	1.3	V
V _{EN-VOU-HYS}	Enable input hysteresis for V _{OUT}	Hysteresis below V _{ENABLE-H} ; falling		110		mV
I _{LKG-EN}	Enable input leakage current	V _{EN} = 3.3V		2.7		nA
INTERNAL LDO (VCC PIN)						
V _{CC}	Internal V _{CC} voltage	6 V ≤ V _{IN} ≤ 65 V	4.75	5	5.25	V
V _{CC-UVLO-Rising}	Internal V _{CC} undervoltage lockout	V _{CC} rising	3.6	3.8	4.0	V
V _{CC-UVLO-Falling}	Internal V _{CC} undervoltage lockout	V _{CC} falling	3.1	3.3	3.5	V
VOLTAGE REFERENCE (FB PIN)						
V _{FB}	Feedback voltage		0.985	1	1.015	V
I _{LKG-FB}	Feedback leakage current	FB = 1 V		2.1		nA
CURRENT LIMITS AND HICUP						
I _{SC}	High-side current limit ⁽³⁾		1.6	2	2.4	A
I _{LS-LIMIT}	Low-side current limit ⁽³⁾		1	1.3	1.6	A
I _{L-ZC}	Zero cross detector threshold	PFM variants only		0.04		A
I _{PEAK-MIN}	Minimum inductor peak current ⁽³⁾			0.28		A

6.5 Electrical Characteristics (continued)

Limits apply over operating junction temperature (T_J) range of -40°C to $+125^{\circ}\text{C}$, unless otherwise stated. Minimum and Maximum limits⁽¹⁾ are specified through test, design or statistical correlation. Typical values represent the most likely parametric norm at $T_J = 25^{\circ}\text{C}$, and are provided for reference purposes only. Unless otherwise stated, the following conditions apply: $V_{IN} = 24\text{ V}$.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER GOOD (PGOOD PIN)						
$V_{PG-HIGH-UP}$	Power-Good upper threshold - rising	% of FB voltage	105%	107%	110%	
$V_{PG-LOW-DN}$	Power-Good lower threshold - falling	% of FB voltage	90%	93%	95%	
V_{PG-HYS}	Power-Good hysteresis (rising & falling)	% of FB voltage		1.5%		
$V_{PG-VALID}$	Minimum input voltage for proper Power-Good function				2	V
R_{PG}	Power-Good on-resistance	$V_{EN} = 2.5\text{ V}$		80	165	Ω
R_{PG}	Power-Good on-resistance	$V_{EN} = 0\text{ V}$		35	90	Ω
MOSFETS						
$R_{DS-ON-HS}$	High-side MOSFET ON-resistance	$I_{OUT} = 0.5\text{ A}$		245	465	m Ω
$R_{DS-ON-LS}$	Low-side MOSFET ON-resistance	$I_{OUT} = 0.5\text{ A}$		165	310	m Ω

- (1) MIN and MAX limits are 100% production tested at 25°C . Limits over the operating temperature range verified through correlation using Statistical Quality Control (SQC) methods. Limits are used to calculate Average Outgoing Quality Level (AOQL).
- (2) This is the current used by the device open loop. This value does not represent the total input current of the system when in regulation.
- (3) The current limit values in this table are tested, open loop, in production. The values can differ from those found in a closed loop application.

6.6 Timing Requirements

Limits apply over operating junction temperature (T_J) range of -40°C to $+125^{\circ}\text{C}$, unless otherwise stated. Minimum and Maximum limits⁽¹⁾ are specified through test, design or statistical correlation. Typical values represent the most likely parametric norm at $T_J = 25^{\circ}\text{C}$, and are provided for reference purposes only. Unless otherwise stated, the following conditions apply: $V_{IN} = 24\text{ V}$.

		MIN	NOM	MAX	UNIT
t_{ON-MIN}	Minimum switch on-time		92		ns
$t_{OFF-MIN}$	Minimum switch off-time		80	102	ns
t_{ON-MAX}	Maximum switch on-time		7	12	μs
t_{SS}	Internal soft-start time	3	4.5	6	ms

- (1) MIN and MAX limits are 100% production tested at 25°C . Limits over the operating temperature range verified through correlation using Statistical Quality Control (SQC) methods. Limits are used to calculate Average Outgoing Quality Level (AOQL).

6.7 Switching Characteristics

$T_J = -40^{\circ}\text{C}$ to 125°C , $V_{IN} = 24\text{ V}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
OSCILLATOR						
F_{OSC}	Internal oscillator frequency	400-kHz variant	340	400	460	kHz

6.8 System Characteristics

The following specifications apply to a typical application circuit with nominal component values. Specifications in the typical (TYP) column apply to $T_J = 25^\circ\text{C}$ only. Specifications in the minimum (MIN) and maximum (MAX) columns apply to the case of typical components over the temperature range of $T_J = -40^\circ\text{C}$ to 125°C . *These specifications are not specified by production testing.*

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{IN}	Operating input voltage range		4.2		65	V
V_{OUT}	Adjustable output voltage regulation ⁽¹⁾	PFM operation	-1.5%		2.5%	
I_{SUPPLY}	Input supply current when in regulation	$V_{IN} = 24\text{ V}$, $V_{OUT} = 3.3\text{ V}$, $I_{OUT} = 0\text{ A}$, $R_{FBT} = 1\text{ M}\Omega$, PFM variant		26		μA
D_{MAX}	Maximum switch duty cycle ⁽²⁾			98%		
V_{HC}	FB pin voltage required to trip short-circuit hiccup mode			0.4		V
t_D	Switch voltage dead time			2		ns
T_{SD}	Thermal shutdown temperature	Shutdown temperature		170		$^\circ\text{C}$
T_{SD}	Thermal shutdown temperature	Recovery temperature		158		$^\circ\text{C}$

(1) Deviation in V_{OUT} from nominal output voltage value at $V_{IN} = 24\text{ V}$, $I_{OUT} = 0\text{ A}$ to full load

(2) In dropout the switching frequency drops to increase the effective duty cycle. The lowest frequency is clamped at approximately: $F_{MIN} = 1 / (t_{ON-MAX} + t_{OFF-MIN})$. $D_{MAX} = t_{ON-MAX} / (t_{ON-MAX} + t_{OFF-MIN})$.

6.9 Typical Characteristics

Unless otherwise specified the following conditions apply: $T_A = 25^\circ\text{C}$ and $V_{IN} = 24\text{ V}$.

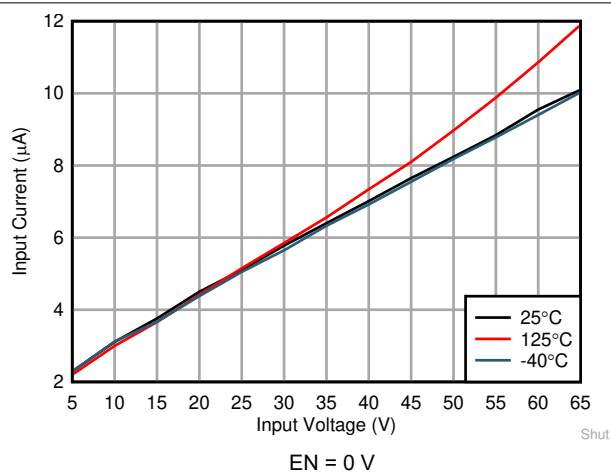


Figure 6-1. Shutdown Supply Current

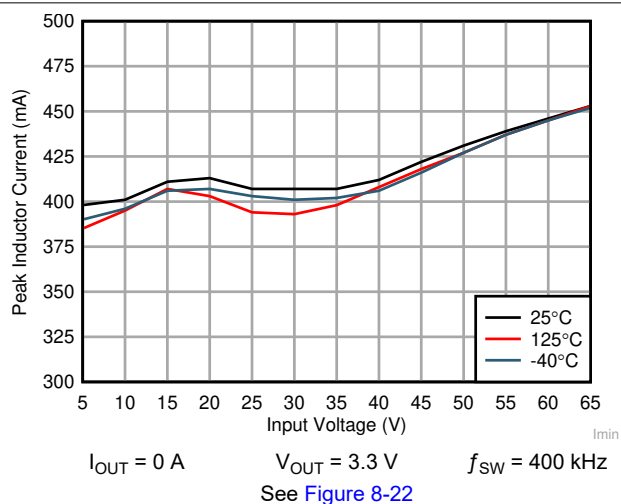


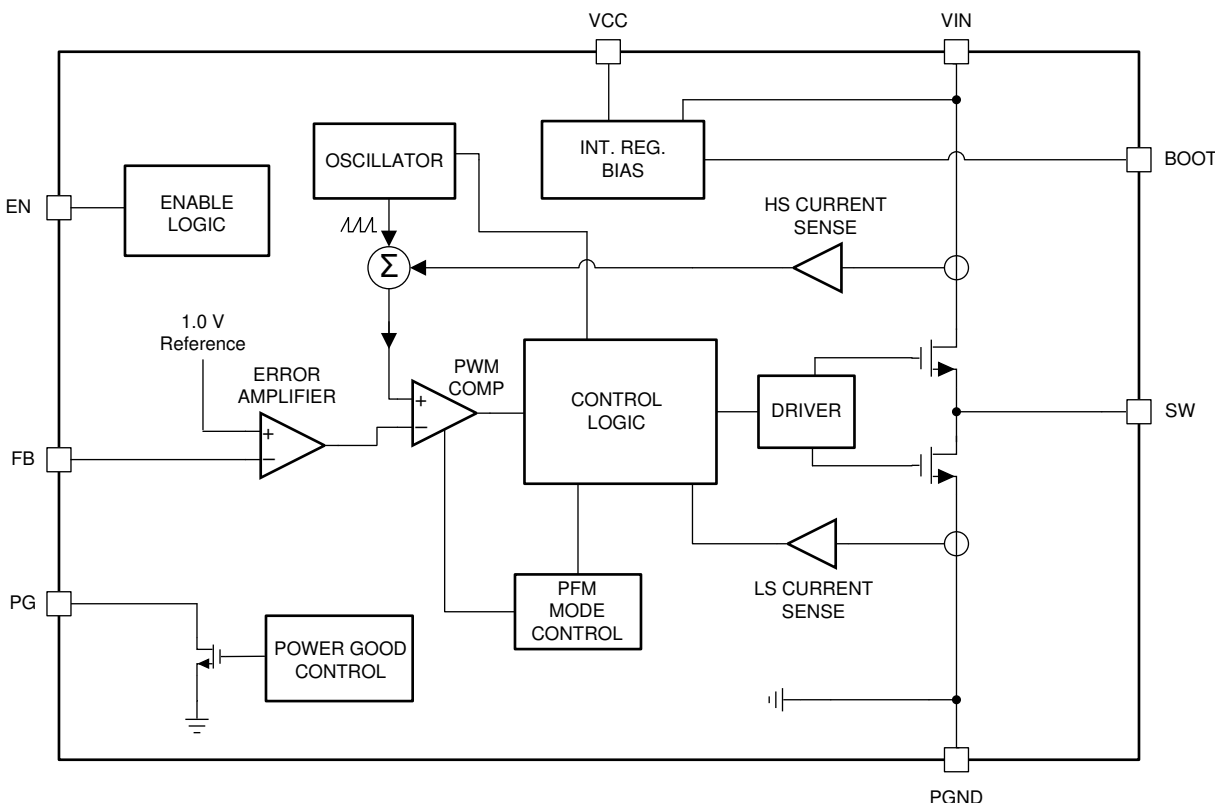
Figure 6-2. $I_{PEAK-MIN}$

7 Detailed Description

7.1 Overview

The LMR36510 is a synchronous peak-current mode buck regulator designed for a wide variety of industrial applications. The regulator automatically switches modes between PFM and PWM, depending on load. At heavy loads, the device operates in PWM at a constant switching frequency. At light loads, the mode changes to PFM with diode emulation, allowing DCM. This reduces the input supply current and keeps efficiency high. The device features internal loop compensation, which reduces design time and requires fewer external components than externally compensated regulators.

7.2 Functional Block Diagram



7.3 Feature Description

7.3.1 Power-Good Flag Output

The power-good flag function (PG output pin) of the LMR36510 can be used to reset a system microprocessor whenever the output voltage is out of regulation. This open-drain output goes low under fault conditions, such as current limit and thermal shutdown, as well as during normal start-up. A glitch filter prevents false flag operation for short excursions of the output voltage, such as during line and load transients. Output voltage excursions lasting less than t_{PG} do not trip the power-good flag. Power-good operation can best be understood by referencing [Figure 7-1](#) and [Figure 7-2](#). Note that during initial power-up, a delay of about 4 ms (typical) is inserted from the time that EN is asserted to the time that the power-good flag goes high. This delay only occurs during start-up and is not encountered during normal operation of the power-good function.

The power-good output consists of an open-drain NMOS, requiring an external pullup resistor to a suitable logic supply. It can also be pulled up to either VCC or V_{OUT} through an appropriate resistor, as desired. If this function is not needed, the PG pin must be grounded. When EN is pulled low, the flag output is also forced low. With EN low, power good remains valid as long as the input voltage is ≥ 2 V (typical). Limit the current into this pin to ≤ 4 mA.

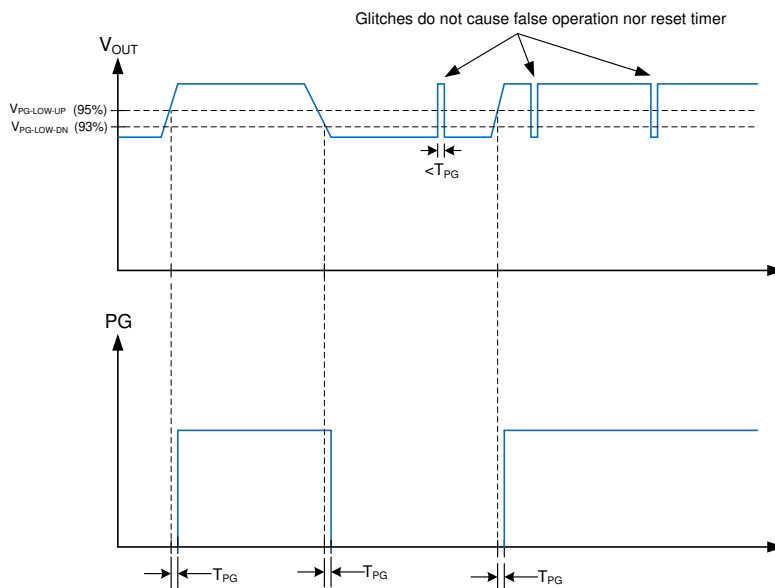


Figure 7-1. Static Power-Good Operation

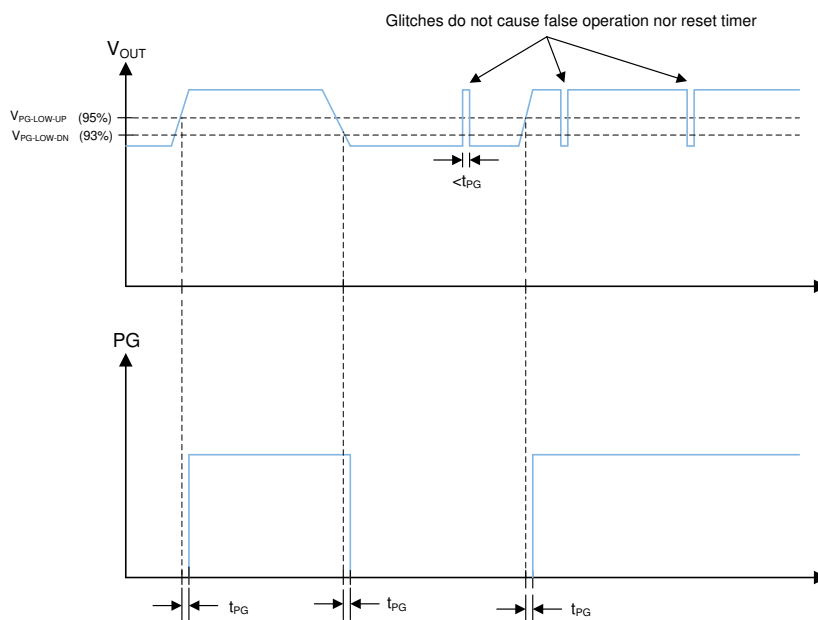


Figure 7-2. Power-Good Timing Behavior

7.3.2 Enable and Start-Up

Start-up and shutdown are controlled by the EN input. This input features precision thresholds, allowing the use of an external voltage divider to provide an adjustable input UVLO (see the [Section 8.2.1.2.9](#) section). Applying a voltage of $\geq V_{EN-VCC_H}$ causes the device to enter standby mode, powering the internal VCC, but not producing an output voltage. Increasing the EN voltage to V_{EN-H} fully enables the device, allowing the device to enter start-up mode and begin the soft-start period. When the EN input is brought below V_{EN-H} by V_{EN-HYS} , the regulator stops running and enters standby mode. If the EN voltage decreases below $V_{EN-VCC-L}$, the device shuts down. [Figure 7-3](#) shows this behavior. The EN input can be connected directly to VIN if this feature is not needed. This input must not be allowed to float. The values for the various EN thresholds can be found in the [Electrical Characteristics](#) table.

The LMR36510 utilizes a reference-based soft start that prevents output voltage overshoots and large inrush currents as the regulator is starting up.

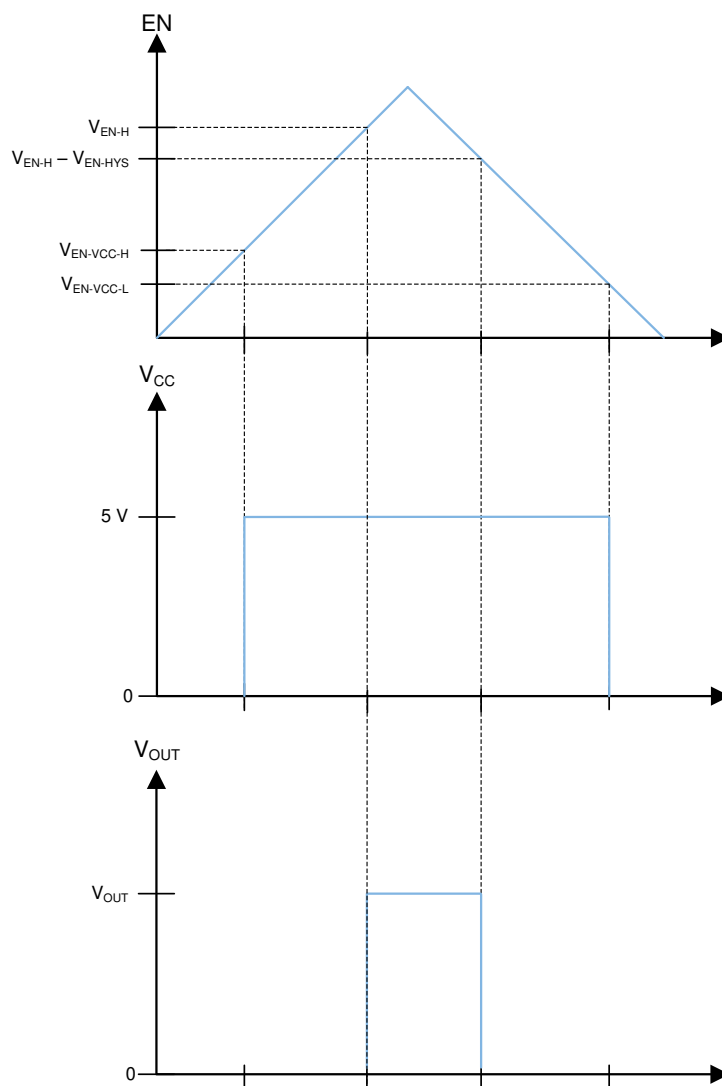


Figure 7-3. Precision Enable Behavior

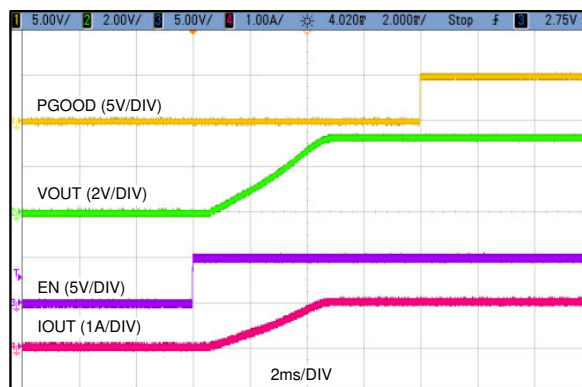


Figure 7-4. Typical Start-up Behavior $V_{IN} = 24\text{ V}$, $V_{OUT} = 5\text{ V}$, $I_{OUT} = 1\text{ A}$

7.3.3 Current Limit and Short Circuit

The LMR36510 incorporates valley current limit for normal overloads and for short-circuit protection. In addition, the high-side power MOSFET is protected from excessive current by a peak-current limit circuit. Cycle-by-cycle current limit is used for overloads while hiccup mode is used for short circuits. Finally, a zero current detector is used on the low-side power MOSFET to implement diode emulation at light loads (see the [Section 9.7](#)).

During overloads, the low-side current limit, I_{LIMIT} , determines the maximum load current that the LMR36510 can supply. When the low-side switch turns on, the inductor current begins to ramp down. If the current does not fall below I_{LIMIT} before the next turnon cycle, then that cycle is skipped, and the low-side MOSFET is left on until the current falls below I_{LIMIT} . This is somewhat different than the more typical peak-current limit and results in [Equation 1](#) for the maximum load current.

$$I_{OUT}|_{max} = I_{LIMIT} + \frac{(V_{IN} - V_{OUT})}{2 \cdot f_{SW} \cdot L} \cdot \frac{V_{OUT}}{V_{IN}} \quad (1)$$

where

- f_{SW} = switching frequency
- L = inductor value

If, during current limit, the voltage on the FB input falls below about 0.4 V due to a short circuit, the device enters into hiccup mode. In this mode, the device stops switching for t_{HC} , or about 94 ms, and then goes through a normal re-start with soft start. If the short-circuit condition remains, the device runs in current limit for about 20 ms (typical) and then shuts down again. This cycle repeats as long as the short-circuit condition persists. This mode of operation reduces the temperature rise of the device during a hard short on the output. Of course, the output current is greatly reduced during hiccup mode. Once the output short is removed and the hiccup delay is passed, the output voltage recovers normally.

The high-side current limit trips when the peak inductor current reaches I_{SC} . This is a cycle-by-cycle current limit and does not produce any frequency or load current foldback. It is meant to protect the high-side MOSFET from excessive current. Under some conditions, such as high input voltages, this current limit can trip before the low-side protection. Under this condition, I_{SC} determines the maximum output current. Note that I_{SC} varies with duty cycle.

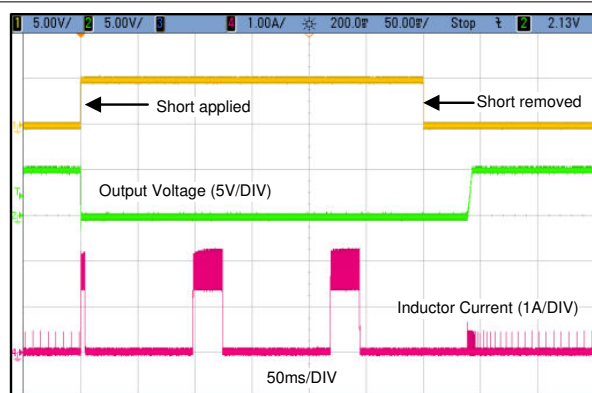


Figure 7-5. Short-Circuit Transient and Recovery

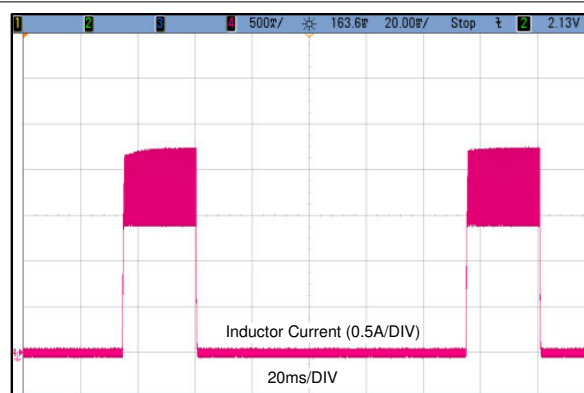


Figure 7-6. Inductor Current Burst in Short Circuit Mode

7.3.4 Undervoltage Lockout and Thermal Shutdown

The LMR36510 incorporates an undervoltage lockout feature on the output of the internal LDO at the VCC pin. When VCC reaches about 3.7 V, the device is ready to receive an EN signal and start up. When VCC falls below about 3 V, the device shuts down, regardless of EN status. Because the LDO is in dropout during these transitions, the previously mentioned values roughly represent the input voltage levels during the transitions.

Thermal shutdown is provided to protect the regulator from excessive junction temperature. When the junction temperature reaches about 170°C, the device shuts down; re-start occurs when the temperature falls to about 158°C. For safe operation, the device must not be allowed to go into a short circuit condition while in thermal shutdown.

7.4 Device Functional Modes

7.4.1 Auto Mode

In auto mode, the device moves between PWM and PFM as the load changes. At light loads, the regulator operates in PFM. At higher loads, the mode changes to PWM.

In PWM, the regulator operates as a constant frequency, current mode, full-synchronous converter using PWM to regulate the output voltage. While operating in this mode, the output voltage is regulated by switching at a constant frequency and modulating the duty cycle to control the power to the load. This provides excellent line and load regulation and low output voltage ripple.

In PFM, the high-side MOSFET is turned on in a burst of one or more pulses to provide energy to the load. The duration of the burst depends on how long the inductor current takes to reach $I_{PEAK-MIN}$. The frequency of these bursts is adjusted to regulate the output, while diode emulation is used to maximize efficiency (see the [Section 9.7](#)). This mode provides high light-load efficiency by reducing the amount of input supply current required to regulate the output voltage at small loads. This trades off very good light-load efficiency for larger output voltage ripple and variable switching frequency. Also, a small increase in output voltage occurs at light loads. The actual switching frequency and output voltage ripple depend on the input voltage, output voltage, and load.

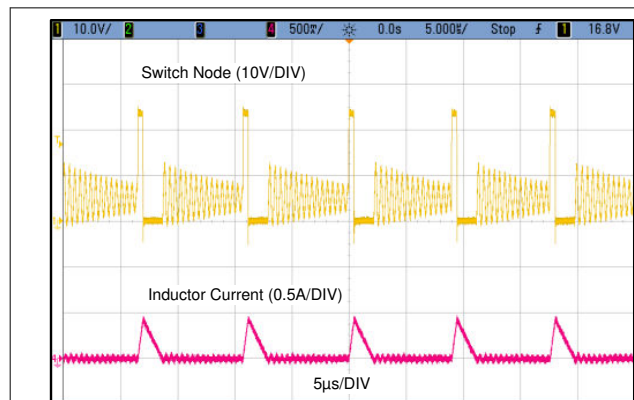


Figure 7-7. Typical PFM Switching Waveforms $V_{IN} = 24\text{ V}$, $V_{OUT} = 5\text{ V}$, $I_{OUT} = 50\text{ mA}$

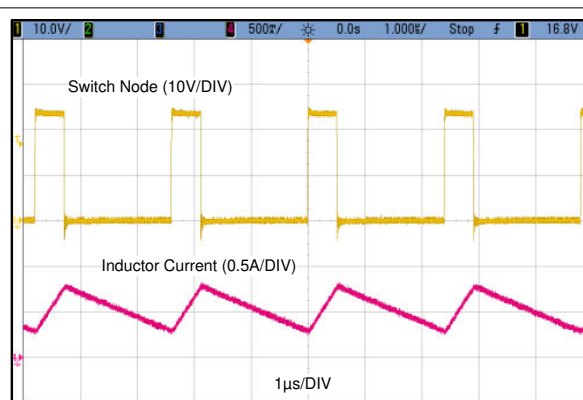


Figure 7-8. Typical PWM Switching Waveforms $V_{IN} = 24\text{ V}$, $V_{OUT} = 5\text{ V}$, $I_{OUT} = 0.5\text{ A}$, $f_S = 400\text{ kHz}$

7.4.2 Dropout

The dropout performance of any buck regulator is affected by the $R_{DS(on)}$ of the power MOSFETs, the DC resistance of the inductor, and the maximum duty cycle that the controller can achieve. As the input voltage is reduced to the output voltage, the off-time of the high-side MOSFET starts to approach the minimum value. Beyond this point, the switching can become erratic and the output voltage falls out of regulation. To avoid this problem, the LMR36510 automatically reduces the switching frequency to increase the effective duty cycle and maintain regulation. In this data sheet, the dropout voltage is defined as the difference between the input and output voltage when the output has dropped by 1% of the nominal value. Under this condition, the switching frequency has dropped to the minimum value of about 140 kHz. Note that the 0.4 V short-circuit detection threshold is not activated when in dropout mode.

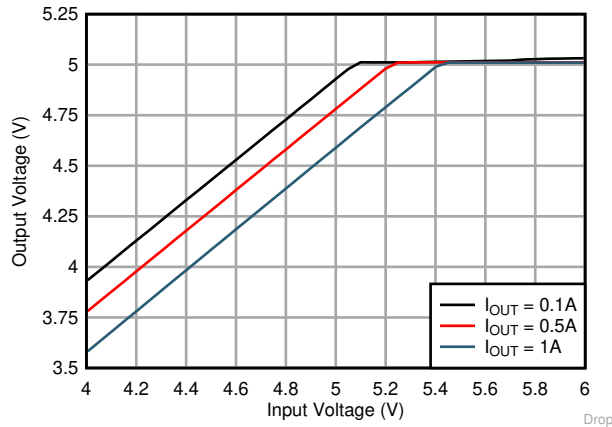


Figure 7-9. Overall Dropout Characteristic $V_{OUT} = 5$ V

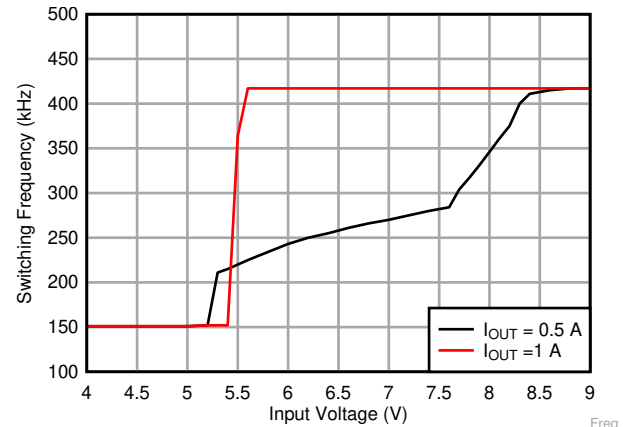


Figure 7-10. Frequency Dropout Characteristics $f_{sw} = 400$ kHz

7.4.3 Minimum Switch On-Time

Every switching regulator has a minimum controllable on-time dictated by the inherent delays and blanking times associated with the control circuits. This imposes a minimum switch duty cycle, therefore, a minimum conversion ratio. The constraint is encountered at high input voltages and low output voltages. To help extend the minimum controllable duty cycle, the LMR36510 automatically reduces the switching frequency when the minimum on-time limit is reached. This way, the converter can regulate the lowest programmable output voltage at the maximum input voltage. An estimate for the approximate input voltage, for a given output voltage, before frequency foldback occurs is found in Equation 2. As the input voltage is increased, the switch on-time (duty cycle) reduces to regulate the output voltage. When the on-time reaches the limit, the switching frequency drops, while the on-time remains fixed.

$$V_{IN} \leq \frac{V_{OUT}}{t_{ON} \cdot f_{SW}} \quad (2)$$

8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

8.1 Application Information

The LMR36510 step-down DC-to-DC converter is typically used to convert a higher DC voltage to a lower DC voltage with a maximum output current of 1 A. The following design procedure can be used to select components for the LMR36510.

Note

All of the capacitance values given in the following application information refer to *effective* values; unless otherwise stated. The *effective* value is defined as the actual capacitance under DC bias and temperature, not the rated or nameplate values. Use high-quality, low-ESR, ceramic capacitors with an X7R or better dielectric throughout. All high value ceramic capacitors have a large voltage coefficient in addition to normal tolerances and temperature effects. Under DC bias, the capacitance drops considerably. Large case sizes and higher voltage ratings are better in this regard. To help mitigate these effects, multiple capacitors can be used in parallel to bring the minimum *effective* capacitance up to the required value. This can also ease the RMS current requirements on a single capacitor. A careful study of bias and temperature variation of any capacitor bank must be made in order to ensure that the minimum value of *effective* capacitance is provided.

8.2 Typical Application

shows a typical application circuit for the LMR36510. This device is designed to function over a wide range of external components and system parameters. However, the internal compensation is optimized for a certain range of external inductance and output capacitance. As a quick-start guide, [Figure 8-1](#) provides typical component values for a range of the most common output voltages.

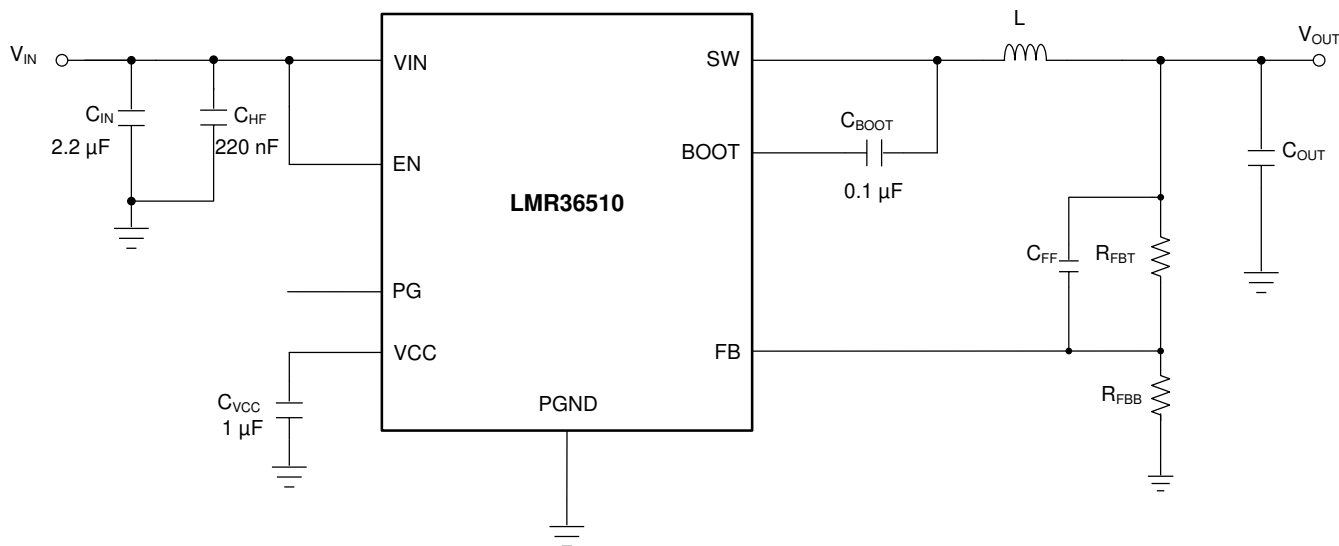


Figure 8-1. Example Applications Circuit

Table 8-1. Typical External Component Values

f_{sw} (kHz)	V_{OUT} (V)	L (μ H)	NOMINAL C_{OUT} (RATED CAPACITANCE) (1)	MINIMUM C_{OUT} (RATED CAPACITANCE) (2)	R_{FBT} (Ω)	R_{FBB} (Ω)	C_{IN}	C_{FF}
400	3.3	22	$3 \times 22 \mu F$	$2 \times 22 \mu F$	100 k	43.2 k	$1 \times 2.2 \mu F + 220 \text{ nF}$	None
400	5	22	$2 \times 22 \mu F$	$1 \times 22 \mu F$	100 k	24.9 k	$1 \times 2.2 \mu F + 220 \text{ nF}$	None
400	12	47	$3 \times 10 \mu F$	$2 \times 10 \mu F$	100 k	9.09 k	$1 \times 2.2 \mu F + 220 \text{ nF}$	None

- (1) Optimized for superior load transient performance from 0 to 100% rated load
(2) Optimized for size constrained end applications

8.2.1 Design 1: Low Power 24-V, 1-A Buck Converter

8.2.1.1 Design Requirements

The following are example requirements for a typical 5-V or 3.3-V application. The input voltages are here for illustration purposes only. See the [Section 6](#) for the operating input voltage range.

Table 8-2. Detailed Design Parameters

DESIGN PARAMETER	EXAMPLE VALUE
Input voltage	12 V to 24 V steady state, 4.2 V to 60-V transients
Output voltage	5 V
Maximum output current	0 A to 1 A
Switching frequency	400 kHz
Current consumption at 0-A load	Critical: Need to ensure low current consumption to reduce battery drain
Switching frequency at 0-A load	Not critical: Need fixed frequency operation at high load only

Table 8-3. List of Components for Design 1

V_{OUT}	FREQUENCY	R_{FBB}	C_{OUT}	L	U1
5 V	400 kHz	24.9 k Ω	$2 \times 22 \mu F$	22 μ H	LMR36510
3.3 V	400 kHz	43.2 k Ω	$3 \times 22 \mu F$	22 μ H	LMR36510

8.2.1.2 Detailed Design Procedure

The following design procedure applies to [Figure 8-1](#) and [Table 8-2](#).

8.2.1.2.1 Choosing the Switching Frequency

The choice of switching frequency is a compromise between conversion efficiency and overall design size. The LMR36510 switching frequency is fixed internal to the IC, therefore, a value of 400 kHz is used in this design.

8.2.1.2.2 Setting the Output Voltage

The output voltage of LMR36510 is externally adjustable using a resistor divider network. The range of recommended output voltage is found in the [Recommended Operating Conditions](#). The divider network is comprised of R_{FBT} and R_{FBB} , and closes the loop between the output voltage and the converter. The converter regulates the output voltage by holding the voltage on the FB pin equal to the internal reference voltage, V_{REF} . The resistance of the divider is a compromise between excessive noise pickup and excessive loading of the output. Smaller values of resistance reduce noise sensitivity but also reduce the light-load efficiency. The recommended value for R_{FBT} is 100 k Ω with a maximum value of 1 M Ω . If a 1 M Ω is selected for R_{FBT} , then a feedforward capacitor must be used across this resistor to provide adequate loop phase margin (see the [Section 8.2.1.2.8](#) section). Once R_{FBT} is selected, use [Equation 3](#) to select R_{FBB} . V_{REF} is nominally 1 V.

$$R_{FBB} = \frac{R_{FBT}}{\left[\frac{V_{OUT}}{V_{REF}} - 1 \right]} \quad (3)$$

For this 5-V example, values are: $R_{FBT} = 100 \text{ k}\Omega$ and $R_{FBB} = 24.9 \text{ k}\Omega$.

8.2.1.2.3 Inductor Selection

The parameters for selecting the inductor are the inductance and saturation current. The inductance is based on the desired peak-to-peak ripple current and is normally chosen to be in the range of 20% to 40% of the maximum output current. Experience shows that the best value for inductor ripple current is 30% of the maximum load current. Note that when selecting the ripple current for applications with much smaller maximum load than the maximum available from the device, use the maximum device current. Equation 4 can be used to determine the value of inductance. The constant K is the percentage of inductor current ripple. This example uses $K = 0.4$ and with input voltage of 24 V, you can calculate an inductance of $L = 24.74 \text{ }\mu\text{H}$. The standard value of 22 μH is selected.

$$L = \frac{(V_{IN} - V_{OUT})}{f_{SW} \cdot K \cdot I_{OUT\max}} \cdot \frac{V_{OUT}}{V_{IN}} \quad (4)$$

Ideally, the saturation current rating of the inductor is at least as large as the high-side switch current limit, I_{SC} . This ensures that the inductor does not saturate even during a short circuit on the output. When the inductor core material saturates, the inductance falls to a very low value, causing the inductor current to rise very rapidly. Although the valley current limit, I_{LIMIT} , is designed to reduce the risk of current runaway, a saturated inductor can cause the current to rise to high values very rapidly. This can lead to component damage. Do not allow the inductor to saturate. Inductors with a ferrite core material have very *hard* saturation characteristics, but usually have lower core losses than powdered iron cores. Powdered iron cores exhibit a *soft* saturation, allowing some relaxation in the current rating of the inductor. However, they have more core losses at frequencies above about 1 MHz. In any case, the inductor saturation current must not be less than the device low-side current limit, I_{LIMIT} . In order to avoid subharmonic oscillation, the inductance value must not be less than that given in Equation 5:

$$L_{MIN} \geq M \cdot \frac{V_{OUT}}{f_{SW}} \quad (5)$$

where

- L_{MIN} = minimum inductance (H)
- $M = 0.625$
- f_{sw} = switching frequency (Hz)

8.2.1.2.4 Output Capacitor Selection

The value of the output capacitor and the respective ESR determine the output voltage ripple and load transient performance. The output capacitor bank is usually limited by the load transient requirements rather than the output voltage ripple. Equation 6 can be used to estimate a lower bound on the total output capacitance, and an upper bound on the ESR that is required to meet a specified load transient.

$$C_{OUT} \geq \frac{\Delta I_{OUT}}{f_{SW} \cdot \Delta V_{OUT} \cdot K} \cdot \left[(1-D) \cdot (1+K) + \frac{K^2}{12} \cdot (2-D) \right]$$

$$ESR \leq \frac{(2+K) \cdot \Delta V_{OUT}}{2 \cdot \Delta I_{OUT} \left[1+K + \frac{K^2}{12} \cdot \left(1 + \frac{1}{(1-D)} \right) \right]}$$

$$D = \frac{V_{OUT}}{V_{IN}} \quad (6)$$

where

- ΔV_{OUT} = output voltage transient
- ΔI_{OUT} = output current transient
- K = ripple factor from [Section 8.2.1.2.3](#)

Once the output capacitor and ESR have been calculated, [Equation 7](#) can be used to check the output voltage ripple.

$$V_r \cong \Delta I_L \cdot \sqrt{ESR^2 + \frac{1}{(8 \cdot f_{SW} \cdot C_{OUT})^2}} \quad (7)$$

where

- V_r = peak-to-peak output voltage ripple

The output capacitor and ESR can then be adjusted to meet both the load transient and output ripple requirements.

In practice, the output capacitor has the most influence on the transient response and loop-phase margin. Load transient testing and bode plots are the best way to validate any given design and must always be completed before the application goes into production. In addition to the required output capacitance, a small ceramic placed on the output can help reduce high frequency noise. Small case size ceramic capacitors in the range of 1 nF to 100 nF can be very helpful in reducing spikes on the output caused by inductor and board parasitics.

Limit the maximum value of total output capacitance to about 10 times the design value, or 1000 μ F, whichever is smaller. Large values of output capacitance can adversely affect the start-up behavior of the regulator as well as the loop stability. If values larger than noted here must be used, then a careful study of start-up at full load and loop stability must be performed.

8.2.1.2.5 Input Capacitor Selection

The ceramic input capacitors provide a low impedance source to the regulator in addition to supplying the ripple current and isolating switching noise from other circuits. A minimum ceramic capacitance of 2.2- μ F is required on the input of the LMR36510. This must be rated for at least the maximum input voltage that the application requires; preferably twice the maximum input voltage. This capacitance can be increased to help reduce input voltage ripple and maintain the input voltage during load transients. In addition, a small case size 220-nF ceramic capacitor must be used at the input, as close as possible to the regulator. This provides a high frequency bypass for the control circuits internal to the device. For this example, a 1 x 2.2- μ F, 100-V, X7R (or better) ceramic capacitor is chosen. The 220 nF must also be rated at 100-V with an X7R dielectric.

Using an electrolytic capacitor on the input in parallel with the ceramics is often desirable. This is especially true if long leads, traces are used to connect the input supply to the regulator. The moderate ESR of this capacitor can help damp any ringing on the input supply caused by the long power leads. The use of this additional capacitor also helps with voltage dips caused by input supplies with unusually high impedance.

Most of the input switching current passes through the ceramic input capacitors. The approximate RMS value of this current can be calculated from [Equation 8](#) and must be checked against the manufacturer's maximum ratings.

$$I_{RMS} \cong \frac{I_{OUT}}{2} \quad (8)$$

8.2.1.2.6 C_{BOOT}

The LMR36510 requires a bootstrap capacitor connected between the BOOT pin and the SW pin. This capacitor stores energy that is used to supply the gate drivers for the power MOSFETs. A high-quality ceramic capacitor of 100 nF and at least 16 V is required.

8.2.1.2.7 VCC

The VCC pin is the output of the internal LDO used to supply the control circuits of the regulator. This output requires a 1-μF, 16-V ceramic capacitor connected from VCC to GND for proper operation. In general, this output must not be loaded with any external circuitry. However, this output can be used to supply the pullup for the power-good function (see the [Section 7.3.1](#) section). A value in the range of 10 kΩ to 100 kΩ is a good choice in this case. The nominal output voltage on VCC is 5 V.

8.2.1.2.8 C_{FF} Selection

In some cases, a feedforward capacitor can be used across R_{FBT} to improve the load transient response or improve the loop-phase margin. This is especially true when values of R_{FBT} > 100 kΩ are used. Large values of R_{FBT}, in combination with the parasitic capacitance at the FB pin, can create a small signal pole that interferes with the loop stability. A C_{FF} can help mitigate this effect. [Equation 9](#) can be used to estimate the value of C_{FF}. The value found with [Equation 9](#) is a starting point; use lower values to determine if any advantage is gained by the using a C_{FF} capacitor. The [Optimizing Transient Response of Internally Compensated DC-DC Converters with Feed-forward Capacitor Application Report](#) is helpful when experimenting with a feedforward capacitor.

$$C_{FF} < \frac{V_{OUT} \cdot C_{OUT}}{120 \cdot R_{FBT} \cdot \sqrt{\frac{V_{REF}}{V_{OUT}}}} \quad (9)$$

8.2.1.2.9 External UVLO

In some cases, an input UVLO level different than that provided internal to the device is needed. This can be accomplished by using the circuit shown in [Figure 8-2](#). The input voltage at which the device turns on is designated V_{ON} and the turnoff voltage is V_{OFF}. First, a value for R_{ENB} is chosen in the range of 10 kΩ to 100 kΩ and then [Equation 10](#) is used to calculate R_{ENT} and V_{OFF}.

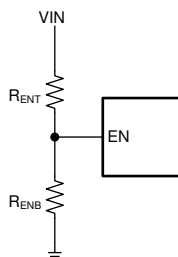


Figure 8-2. Setup for External UVLO Application

$$R_{ENT} = \left(\frac{V_{ON}}{V_{EN-H}} - 1 \right) \cdot R_{ENB}$$

$$V_{OFF} = V_{ON} \cdot \left(1 - \frac{V_{EN-HYS}}{V_{EN}} \right) \quad (10)$$

where

- V_{ON} = V_{IN} turnon voltage
- V_{OFF} = V_{IN} turnoff voltage

8.2.1.2.10 Maximum Ambient Temperature

As with any power conversion device, the LMR36510 dissipates internal power while operating. The effect of this power dissipation is to raise the internal temperature of the converter above ambient. The internal die temperature (T_J) is a function of the ambient temperature, the power loss, and the effective thermal resistance, R_{θJA}, of the device and PCB combination. The maximum internal die temperature for the LMR36510 must be

limited to 150°C. This establishes a limit on the maximum device power dissipation and, therefore, the load current. Equation 11 shows the relationships between the important parameters. Seeing that larger ambient temperatures (T_A) and larger values of $R_{\theta JA}$ reduce the maximum available output current is easy. The converter efficiency can be estimated using the curves provided in this data sheet. If the desired operating conditions cannot be found in one of the curves, then interpolation can be used to estimate the efficiency. Alternatively, the EVM can be adjusted to match the desired application requirements and the efficiency can be measured directly. The correct value of $R_{\theta JA}$ is more difficult to estimate. As stated in the [Semiconductor and IC Package Thermal Metrics Application Report](#), the values given in [Thermal Information](#) are not valid for design purposes and must not be used to estimate the thermal performance of the application. The values reported in that table are measured under a specific set of conditions that are rarely obtained in an actual application.

$$I_{OUT}|_{MAX} = \frac{(T_J - T_A)}{R_{\theta JA}} \cdot \frac{\eta}{(1 - \eta)} \cdot \frac{1}{V_{OUT}} \quad (11)$$

where

- η = efficiency

The effective $R_{\theta JA}$ is a critical parameter and depends on many factors such as the following:

- Power dissipation
- Air temperature/flow
- PCB area
- Copper heat-sink area
- Number of thermal vias under the package
- Adjacent component placement

A typical example of $R_{\theta JA}$ versus copper board area can be found in [Figure 8-3](#). Note that the data given in this graph is for illustration purposes only, and the actual performance in any given application depends on all of the factors mentioned above.

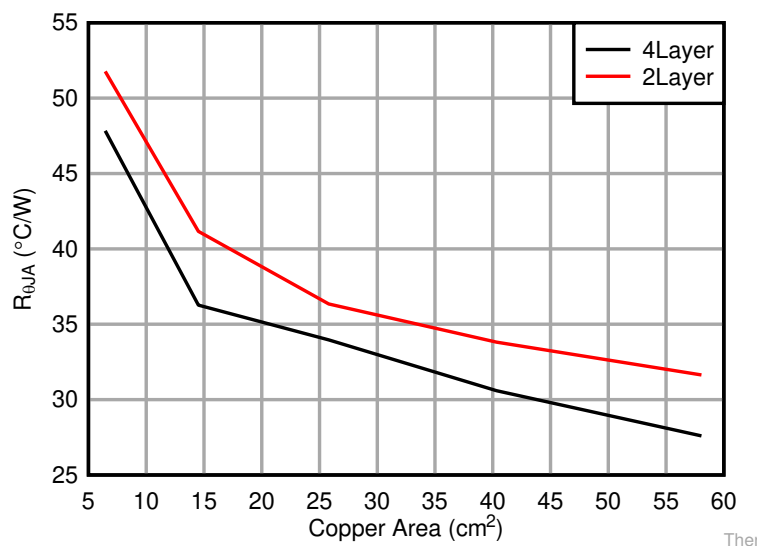


Figure 8-3. $R_{\theta JA}$ versus Copper Board Area

Use the following resources as guides to optimal thermal PCB design and estimate $R_{\theta JA}$ for a given application environment:

- [Thermal Design by Insight not Hindsight Application Report](#)
- [Semiconductor and IC Package Thermal Metrics Application Report](#)
- [Thermal Design Made Simple with LM43603 and LM43602 Application Report](#)
- [Using New Thermal Metrics Application Report](#)

8.2.1.3 Application Curves

Unless otherwise specified, the following conditions apply: $V_{IN} = 24\text{ V}$, $T_A = 25^\circ\text{C}$. Figure 8-22 shows the circuit with the appropriate BOM from Table 8-4.

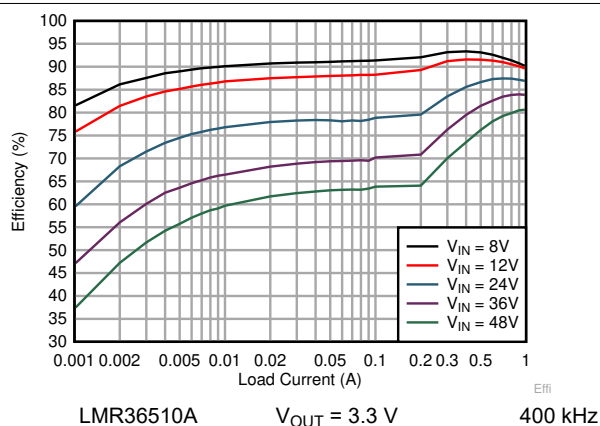


Figure 8-4. Efficiency

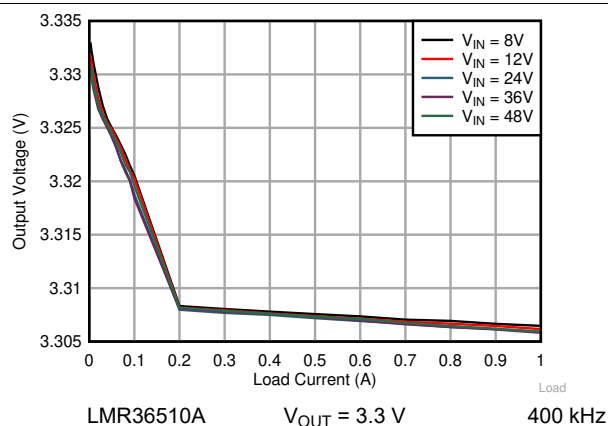


Figure 8-5. Load Regulation

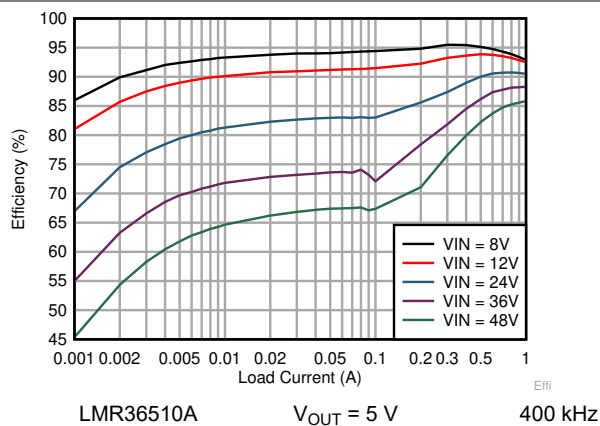


Figure 8-6. Efficiency

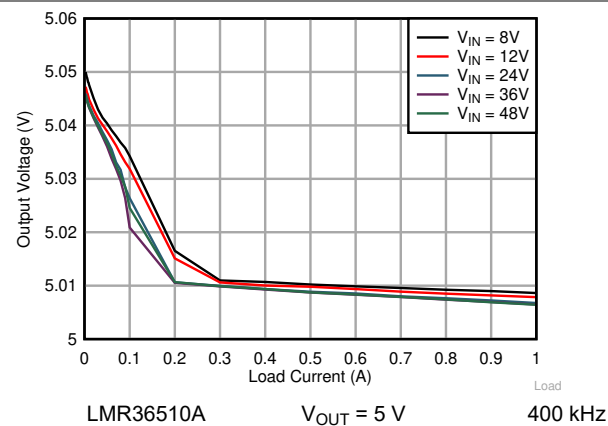


Figure 8-7. Load Regulation

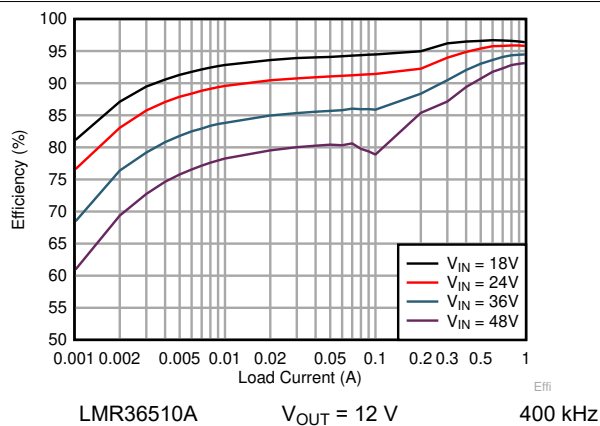


Figure 8-8. Efficiency

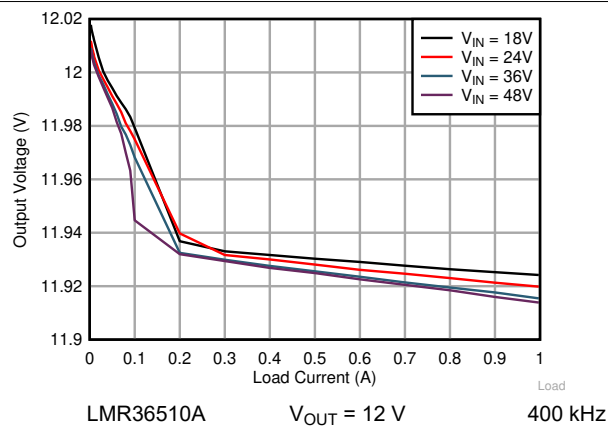


Figure 8-9. Load Regulation

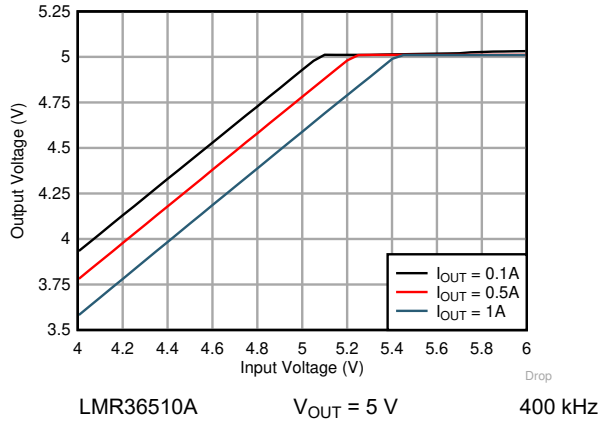


Figure 8-10. Dropout Characteristic

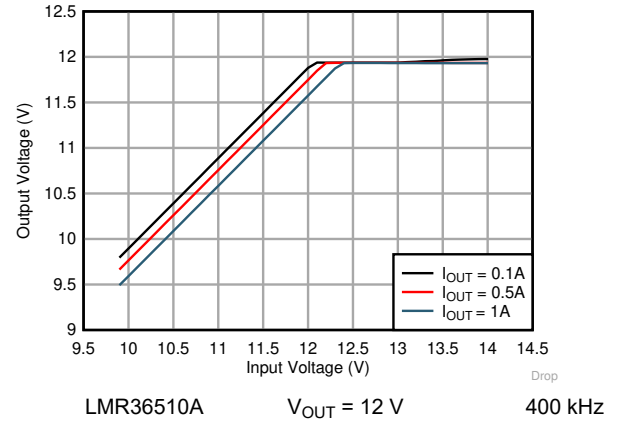


Figure 8-11. Dropout Characteristic

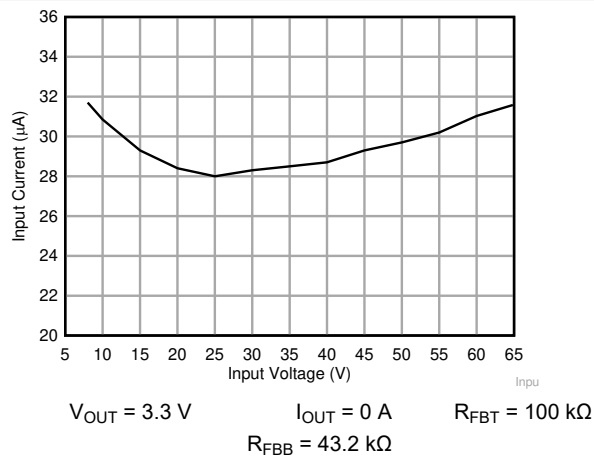


Figure 8-12. Input Supply Current

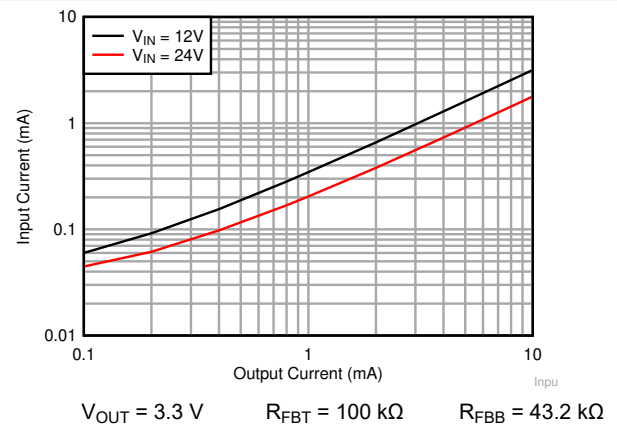


Figure 8-13. Input Supply Current versus Load Current

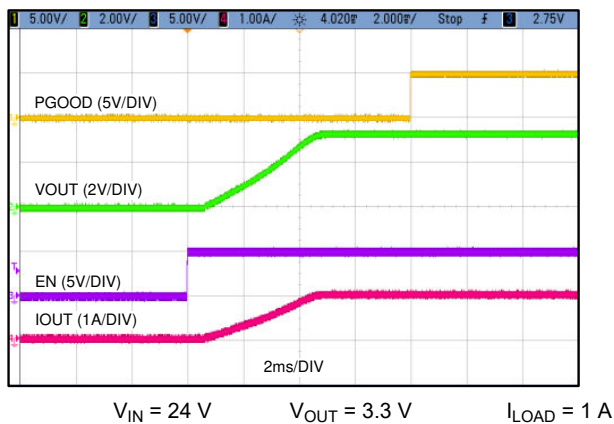


Figure 8-14. Enable Start-Up Waveform

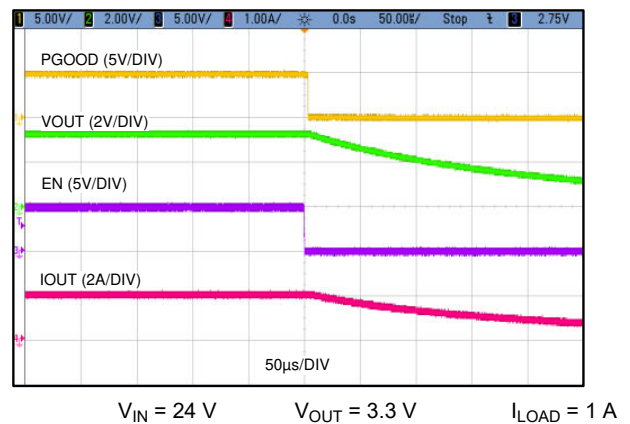
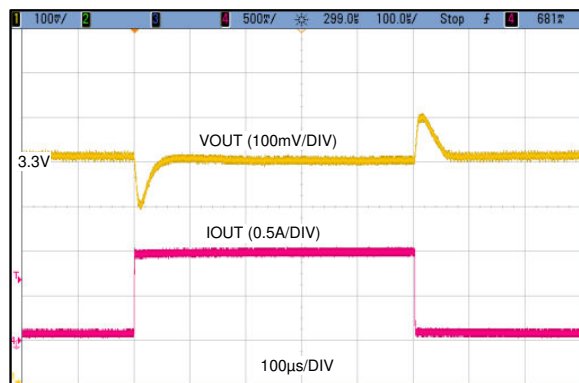
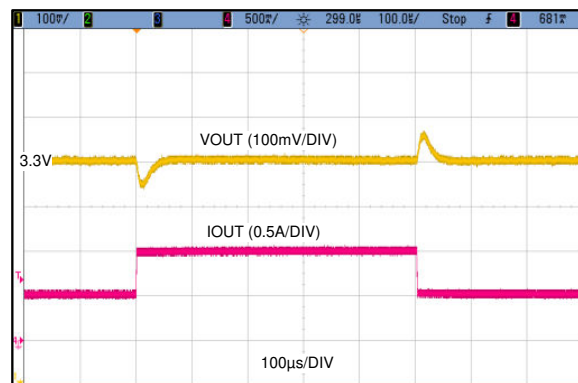


Figure 8-15. Enable Shutdown Waveform



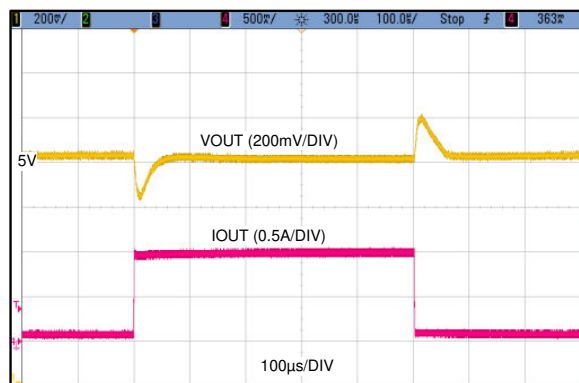
$V_{OUT} = 3.3\text{ V}$ 400 kHz $I_{LOAD} = 0.1\text{ A} - 1\text{ A}$
Slew Rate = 1 A/µs

Figure 8-16. Load Transient



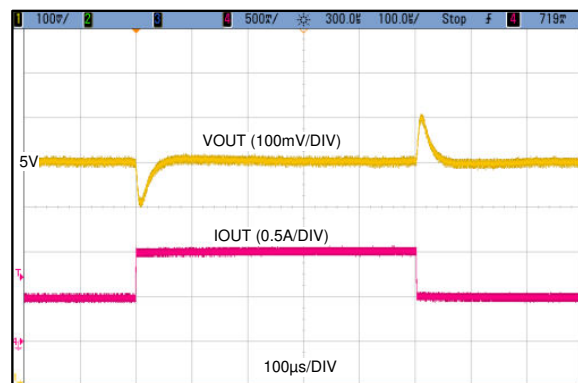
$V_{OUT} = 3.3\text{ V}$ 400 kHz $I_{LOAD} = 0.5\text{ A} - 1\text{ A}$
Slew Rate = 1 A/µs

Figure 8-17. Load Transient



$V_{OUT} = 5\text{ V}$ 400 kHz $I_{LOAD} = 0.1\text{ A} - 1\text{ A}$
Slew Rate = 1 A/µs

Figure 8-18. Load Transient



$V_{OUT} = 5\text{ V}$ 400 kHz $I_{LOAD} = 0.5\text{ A} - 1\text{ A}$
Slew Rate = 1 A/µs

Figure 8-19. Load Transient

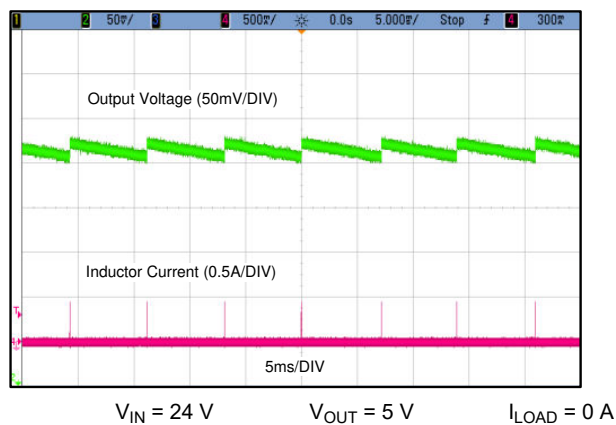


Figure 8-20. Output Ripple at No load

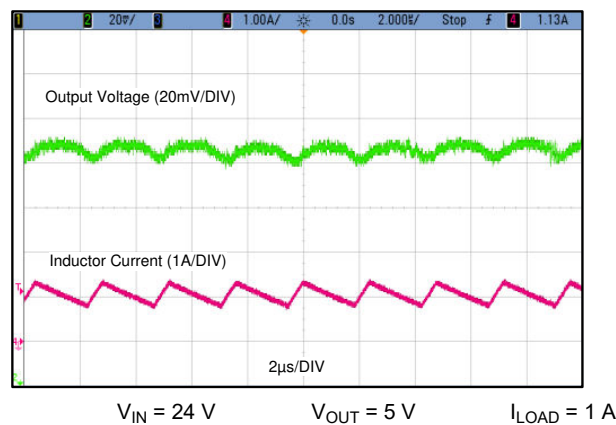


Figure 8-21. Output Ripple at 1 A

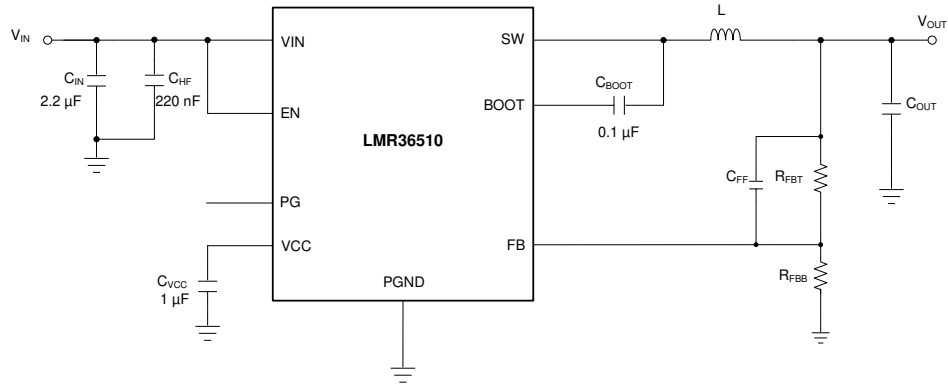


Figure 8-22. Circuit for Typical Application Curves

Table 8-4. BOM for Typical Application Curves

V _{OUT}	FREQUENCY	L	C _{OUT}	R _{FBT}	R _{FBB}	C _{FF}	IC
3.3 V	400 kHz	22 μH, 99.65 mΩ	3 x 22 μF, 25V	46.4 kΩ	20.0 kΩ	None	LMR36510A
5 V	400 kHz	22 μH, 99.65 mΩ	2 x 22 μF, 25V	34.0 kΩ + 46.4 kΩ	20.0 kΩ	None	LMR36510A
12 V	400 kHz	47 μH, 100 mΩ	3 x 22 μF, 25V	100 kΩ	9.09 kΩ	None	LMR36510A

8.3 Best Design Practices

- Do not allow the EN input to float.
- Do not allow the output voltage to exceed the input voltage, nor go below ground.
- Do not use the thermal data given in the [Thermal Information](#) table to design your application.

8.4 Power Supply Recommendations

The characteristics of the input supply must be compatible with the [Section 6](#) found in this data sheet. In addition, the input supply must be capable of delivering the required input current to the loaded regulator. The average input current can be estimated with [Equation 12](#).

$$I_{IN} = \frac{V_{OUT} \cdot I_{OUT}}{V_{IN} \cdot \eta} \quad (12)$$

where

- η is the efficiency

If the regulator is connected to the input supply through long wires or PCB traces, special care is required to achieve good performance. The parasitic inductance and resistance of the input cables can have an adverse effect on the operation of the regulator. The parasitic inductance, in combination with the low-ESR, ceramic input capacitors, can form an underdamped resonant circuit, resulting in overvoltage transients at the input to the regulator. The parasitic resistance can cause the voltage at the VIN pin to dip whenever a load transient is applied to the output. If the application is operating close to the minimum input voltage, this dip can cause the regulator to momentarily shutdown and reset. The best way to solve these kind of issues is to reduce the distance from the input supply to the regulator and use an aluminum or tantalum input capacitor in parallel with the ceramics. The moderate ESR of these types of capacitors help damp the input resonant circuit and reduce any overshoots. A value in the range of 20 μ F to 100 μ F is usually sufficient to provide input damping and help hold the input voltage steady during large load transients.

Sometimes, for other system considerations, an input filter is used in front of the regulator. This can lead to instability, as well as some of the effects mentioned above, unless designed carefully. The [AN-2162 Simple Success With Conducted EMI From DC/DC Converters User's Guide](#) provides helpful suggestions when designing an input filter for any switching regulator.

In some cases, a transient voltage suppressor (TVS) is used on the input of regulators. One class of this device has a *snap-back* characteristic (thyristor type). The use of a device with this type of characteristic is not recommended. When the TVS fires, the clamping voltage falls to a very low value. If this voltage is less than the output voltage of the regulator, the output capacitors discharge through the device and back into the input. This uncontrolled current flow can damage the device.

8.5 Layout

8.5.1 Layout Guidelines

The PCB layout of any DC/DC converter is critical to the optimal performance of the design. Poor PCB layout can disrupt the operation of an otherwise good schematic design. Even if the converter regulates correctly, bad PCB layout can mean the difference between a robust design and one that cannot be mass produced. Furthermore, to a great extent, the EMI performance of the regulator is dependent on the PCB layout. In a buck converter, the most critical PCB feature is the loop formed by the input capacitors and power ground, as shown in [Figure 8-23](#). This loop carries large transient currents that can cause large transient voltages when reacting with the trace inductance. These unwanted transient voltages disrupt the proper operation of the converter. Because of this, the traces in this loop must be wide and short, and the loop area as small as possible to reduce the parasitic inductance. [Figure 8-24](#) shows a recommended layout for the critical components of the LMR36510.

1. *Place the input capacitors as close as possible to the VIN and GND terminals.* VIN and GND pins are adjacent, simplifying the input capacitor placement.
2. *Place the bypass capacitor for VCC close to the VCC pin.* This capacitor must be placed close to the device and routed with short, wide traces to the VCC and GND pins.
3. *Use wide traces for the C_{BOOT} capacitor.* Place C_{BOOT} close to the device with short, wide traces to the BOOT and SW pins. Route the SW pin to the N/C pin and use it to connect the BOOT capacitor to SW.
4. *Place the feedback divider as close as possible to the FB pin of the device.* Place R_{FBB}, R_{FBT}, and C_{FF}, if used, physically close to the device. The connections to FB and GND must be short and close to those

pins on the device. The connection to V_{OUT} can be somewhat longer. However, this latter trace must not be routed near any noise source (such as the SW node) that can capacitively couple into the feedback path of the regulator.

5. *Use at least one ground plane in one of the middle layers.* This plane acts as a noise shield and also act as a heat dissipation path.
6. *Provide wide paths for VIN, VOUT, and GND.* Making these paths as wide and direct as possible reduces any voltage drops on the input or output paths of the converter and maximizes efficiency.
7. *Provide enough PCB area for proper heat-sinking.* As stated in the [Section 8.2.1.2.10](#) section, enough copper area must be used to make sure of a low $R_{\theta JA}$, commensurate with the maximum load current and ambient temperature. The top and bottom PCB layers must be made with two ounce copper; and no less than one ounce. If the PCB design uses multiple copper layers (recommended), these thermal vias can also be connected to the inner layer heat-spreading ground planes.
8. *Keep the switch area small.* Keep the copper area connecting the SW pin to the inductor as short and wide as possible. At the same time, the total area of this node must be minimized to help reduce radiated EMI.

See the following PCB layout resources for additional important guidelines:

- [Layout Guidelines for Switching Power Supplies Application Report](#)
- [Simple Switcher PCB Layout Guidelines Application Report](#)
- [Constructing Your Power Supply- Layout Considerations Seminar](#)
- [Low Radiated EMI Layout Made Simple with LM4360x and LM4600x Application Report](#)

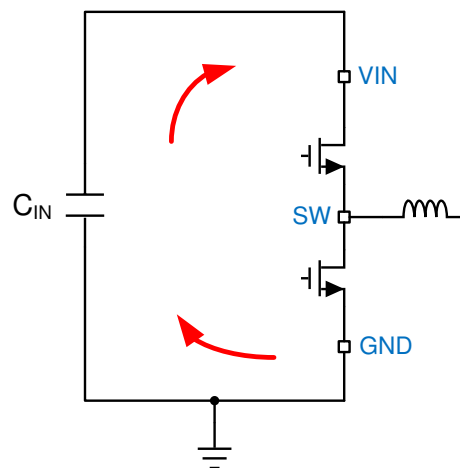


Figure 8-23. Current Loops With Fast Edges

8.5.1.1 Ground and Thermal Considerations

As previously mentioned, TI recommends using one of the middle layers as a solid ground plane. A ground plane provides shielding for sensitive circuits and traces as well as a quiet reference potential for the control circuitry. Connect the PGND pin to the ground planes using vias next to the bypass capacitors. The PGND pin is connected directly to the source of the low-side MOSFET switch and is also connected directly to the grounds of the input and output capacitors. The PGND net contains noise at the switching frequency and can bounce due to load variations. The PGND trace, as well as the VIN and SW traces, must be constrained to one side of the ground planes. The other side of the ground plane contains much less noise; use for sensitive routes.

Use as much copper as possible for the system ground plane, on the top and bottom layers for the best heat dissipation. Use a four-layer board with the copper thickness for the four layers, starting from the top as: 2 oz / 1 oz / 1 oz / 2 oz. A four-layer board with enough copper thickness, and proper layout, provides low current conduction impedance, proper shielding, and lower thermal resistance.

8.5.2 Layout Example

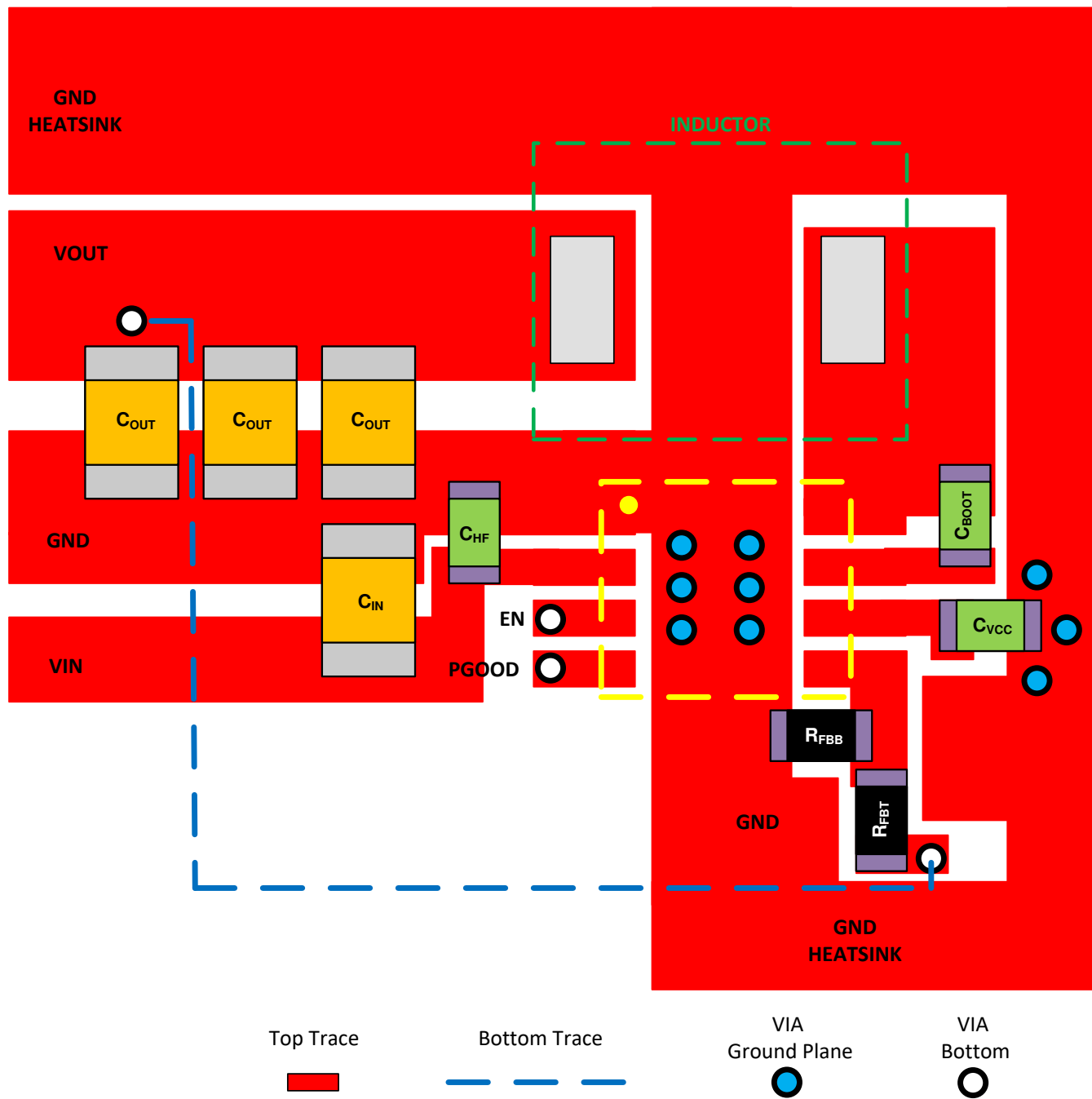


Figure 8-24. Example Layout

9 Device and Documentation Support

9.1 Device Support

9.1.1 Development Support

- [How a DC/DC Converter Package and Pinout Design Can Enhance Automotive EMI Performance Blog](#)
- [Introduction to Buck Converters Features: UVLO, Enable, Soft Start, Power Good Training Video](#)
- [Introduction to Buck Converters: Understanding Mode Transitions Training Video](#)
- [Introduction to Buck Converters: Minimum On-time and Minimum Off-time Operation Training Video](#)
- [Introduction to Buck Converters: Understanding Quiescent Current Specifications Training Video](#)
- [Trade-offs Between Thermal Performance and Small Solution Size with DC/DC Converters Training Video](#)
- [Reduce EMI and Shrink Solution Size with Hot Rod Packaging Training Video](#)

9.2 Documentation Support

9.2.1 Related Documentation

For related documentation see the following:

- Texas Instruments, [Designing High-Performance, Low-EMI Automotive Power Supplies Application Report](#)
- Texas Instruments, [Simple Switcher PCB Layout Guidelines Application Report](#)
- Texas Instruments, [Constructing Your Power Supply- Layout Considerations Seminar](#)
- Texas Instruments, [Low Radiated EMI Layout Made Simple with LM4360x and LM4600x Report](#)
- Texas Instruments, [Semiconductor and IC Package Thermal Metrics Application Report](#)
- Texas Instruments, [Thermal Design Made Simple with LM43603 and LM43602 Application Report](#)
- Texas Instruments, [AN-2162 Simple Success With Conducted EMI From DC/DC Converters User's Guide](#)
- Texas Instruments, [Thermal Design by Insight not Hindsight Application Report](#)
- Texas Instruments, [Semiconductor and IC Package Thermal Metrics Application Report](#)
- Texas Instruments, [Thermal Design Made Simple with LM43603 and LM43602 Application Report](#)
- Texas Instruments, [Using New Thermal Metrics Application Report](#)
- Texas Instruments, [Optimizing Transient Response of Internally Compensated DC-DC Converters with Feed-forward Capacitor Application Report](#)

9.3 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on [ti.com](#). Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

9.4 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

9.5 Trademarks

TI E2E™ is a trademark of Texas Instruments.

SIMPLE SWITCHER® is a registered trademark of Texas Instruments.

All trademarks are the property of their respective owners.

9.6 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

9.7 Glossary

TI Glossary

This glossary lists and explains terms, acronyms, and definitions.

10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

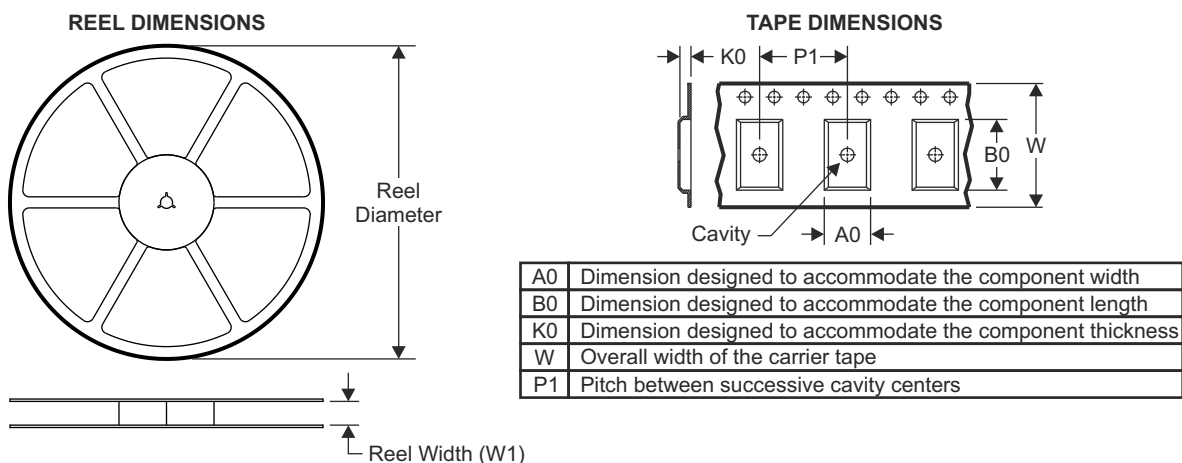
Changes from Revision A (February 2020) to Revision B (November 2025)	Page
• Updated the numbering format for tables, figures, and cross-references throughout the document.....	1
• Added the Device Comparison Table	3

Changes from Revision * (October 2019) to Revision A (February 2020)	Page
• Changed device status from Advance Information to Production Data	1

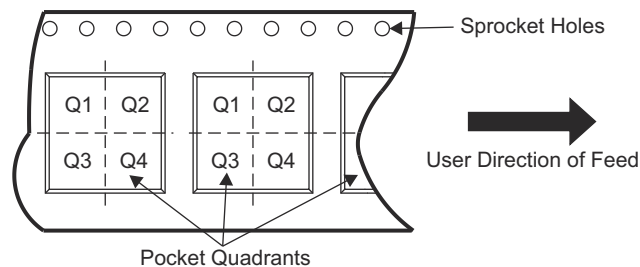
11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

11.1 Tape and Reel Information

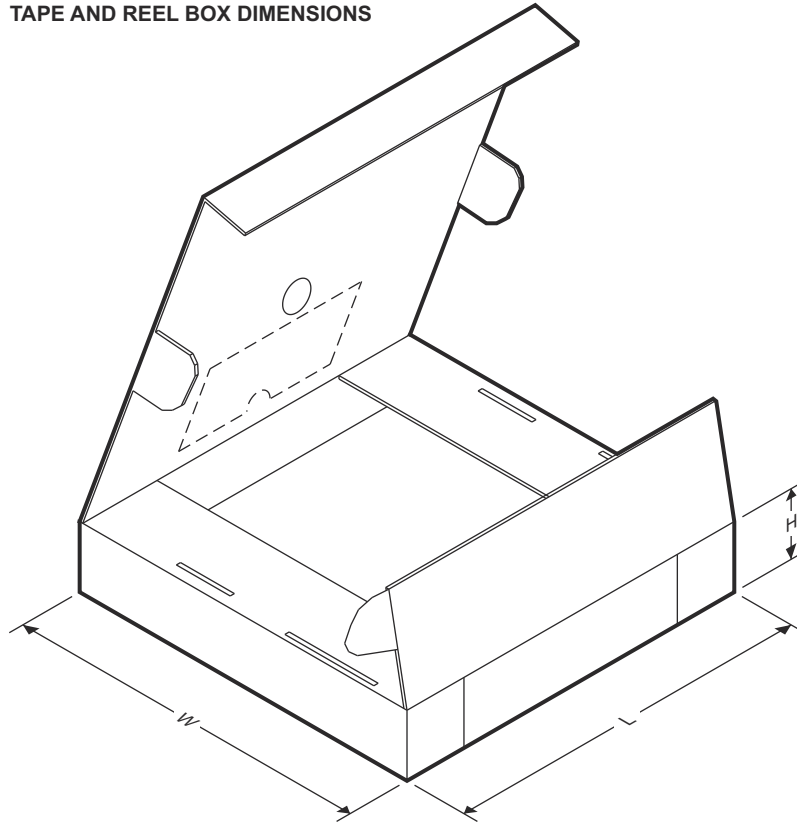


QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
LMR36510ADDAR	SO PowerPAD	DDA	8	2500	330	12.4	6.4	5.2	2.1	8	12	Q1
LMR36510FADDAR	SO PowerPAD	DDA	8	2500	330	12.4	6.4	5.2	2.1	8	12	Q1

TAPE AND REEL BOX DIMENSIONS



Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
LMR36510ADDAR	SO PowerPAD	DDA	8	2500	353	353	36
LMR36510FADDAR	SO PowerPAD	DDA	8	2500	353	353	36

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
LMR36510ADDAR	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU NIPDAUAG	Level-2-260C-1 YEAR	-40 to 125	36510A
LMR36510ADDAR.A	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	36510A
LMR36510FADDAR	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU NIPDAUAG	Level-2-260C-1 YEAR	-40 to 125	36510F
LMR36510FADDAR.A	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	36510F

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

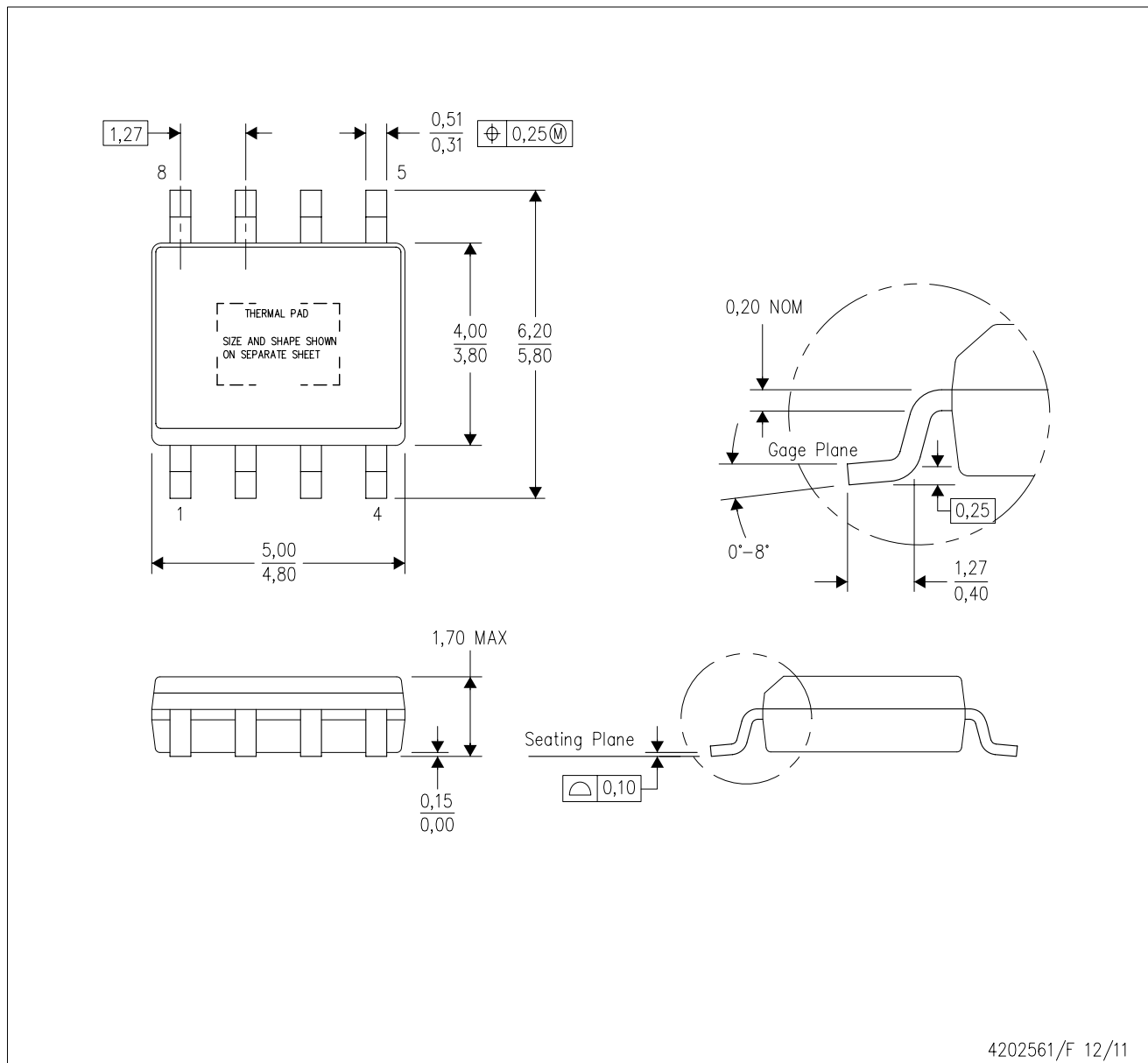
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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DDA (R-PDSO-G8)

PowerPAD™ PLASTIC SMALL-OUTLINE



- NOTES:
- A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5-1994.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion not to exceed 0,15.
 - D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <<http://www.ti.com>>.
 - E. See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
 - F. This package complies to JEDEC MS-012 variation BA

PowerPAD is a trademark of Texas Instruments.

DDA (R-PDSO-G8)

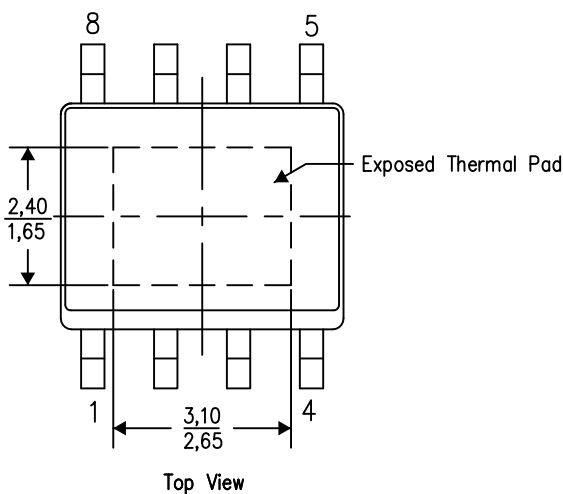
PowerPAD™ PLASTIC SMALL OUTLINE

THERMAL INFORMATION

This PowerPAD™ package incorporates an exposed thermal pad that is designed to be attached to a printed circuit board (PCB). The thermal pad must be soldered directly to the PCB. After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the PowerPAD package and how to take advantage of its heat dissipating abilities, refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 and Application Brief, PowerPAD Made Easy, Texas Instruments Literature No. SLMA004. Both documents are available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.



Exposed Thermal Pad Dimensions

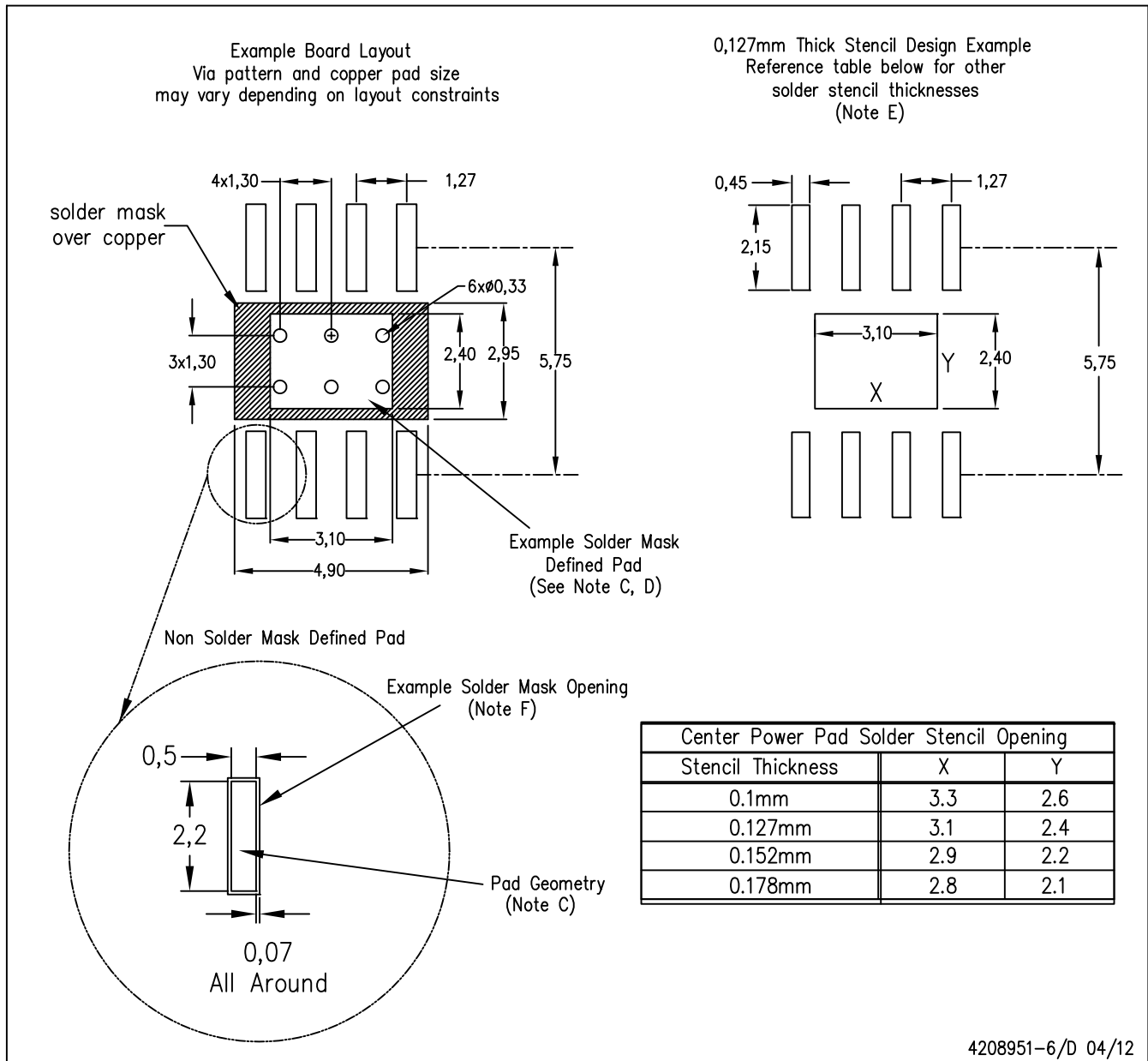
4206322-6/L 05/12

NOTE: A. All linear dimensions are in millimeters

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DDA (R-PDSO-G8)

PowerPAD™ PLASTIC SMALL OUTLINE



4208951-6/D 04/12

- NOTES:
- All linear dimensions are in millimeters.
 - This drawing is subject to change without notice.
 - Publication IPC-7351 is recommended for alternate designs.
 - This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002, SLMA004, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <<http://www.ti.com>>. Publication IPC-7351 is recommended for alternate designs.
 - Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Example stencil design based on a 50% volumetric metal load solder paste. Refer to IPC-7525 for other stencil recommendations.
 - Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

PowerPAD is a trademark of Texas Instruments.

PowerPAD™ SOIC - 1.7 mm max height

Technical drawing of a microelectronic package showing top, side, and detail views with dimensions and callouts.

Top View:

- Overall width: 6.2 TYP, 5.8
- Overall height: 5.0, 4.8, NOTE 3
- Pin 1 ID AREA (hatched)
- Pin 1 location: 1
- Pin 8 location: 8
- Pin 5 location: 5
- Pin 4 location: 4
- Pin 3 location: 3
- Pin 2 location: 2
- Pin 7 location: 7
- Pin 6 location: 6
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- Pin 246 location: 246
- Pin 2

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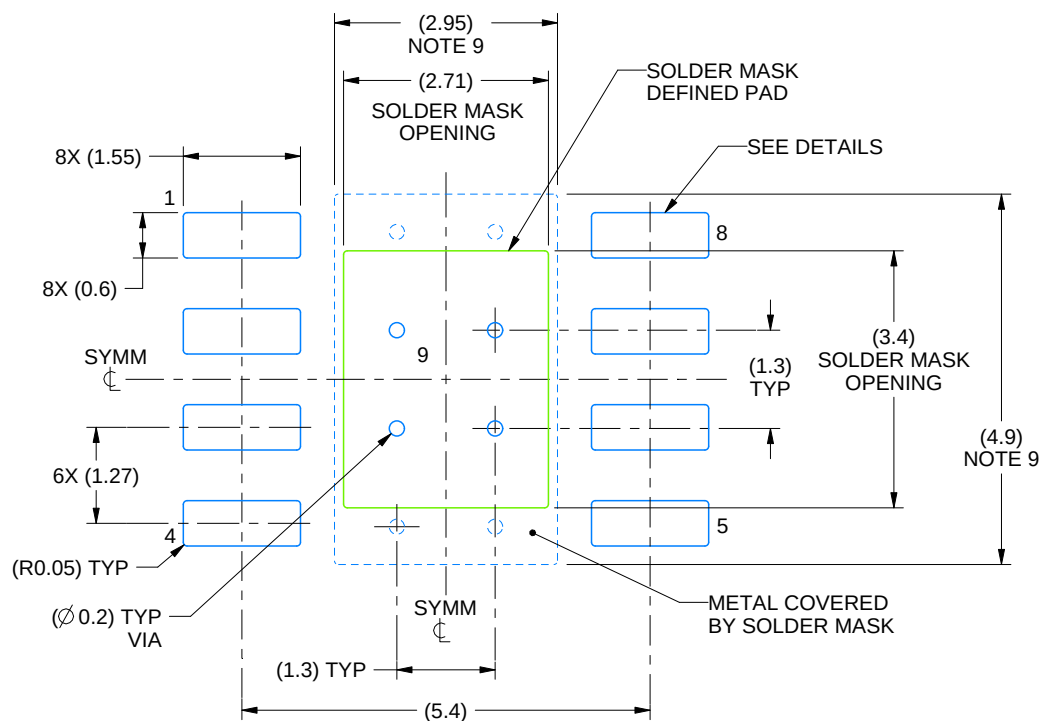
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MS-012.

EXAMPLE BOARD LAYOUT

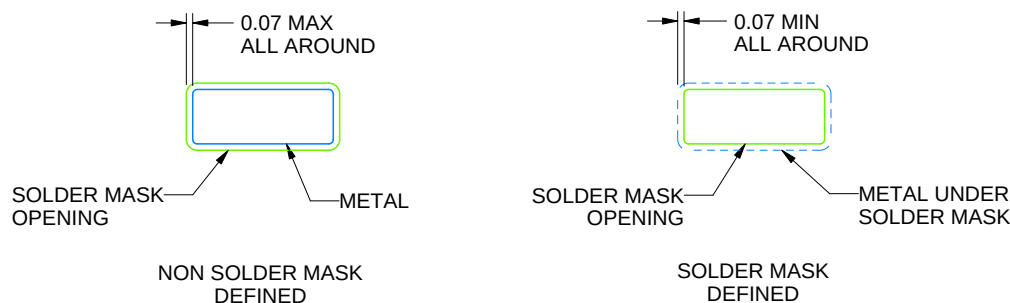
DDA0008B

PowerPAD™ SOIC - 1.7 mm max height

PLASTIC SMALL OUTLINE



LAND PATTERN EXAMPLE
SCALE:10X



SOLDER MASK DETAILS
PADS 1-8

4214849/B 09/2025

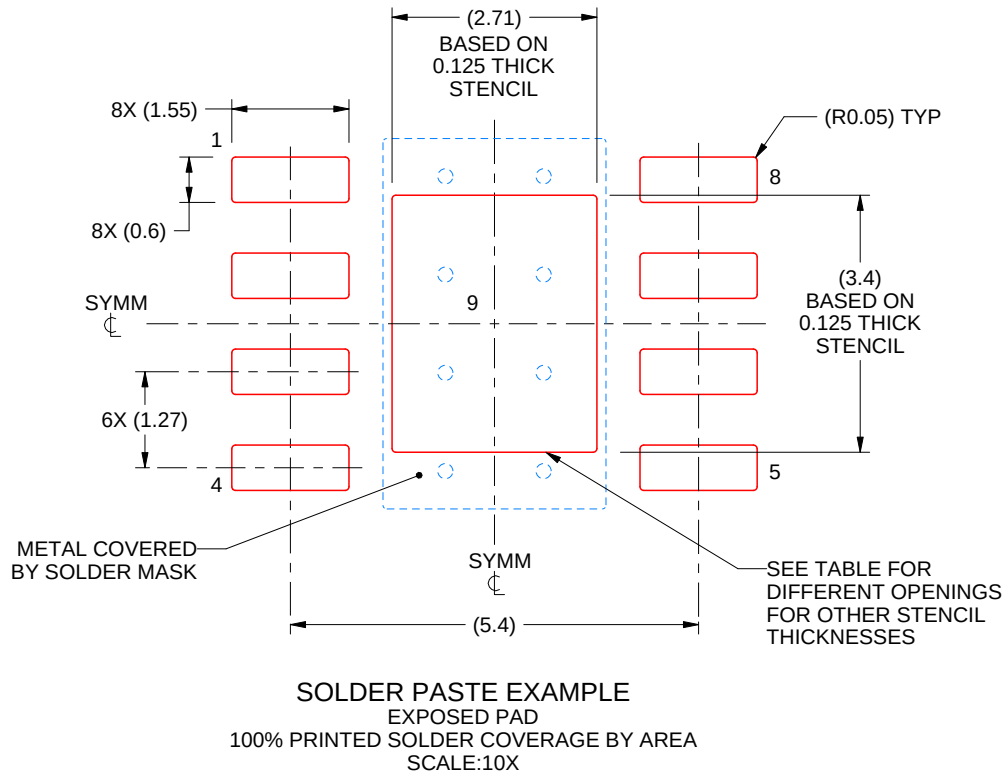
NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature numbers SLMA002 (www.ti.com/lit/slma002) and SLMA004 (www.ti.com/lit/slma004).
9. Size of metal pad may vary due to creepage requirement.
10. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

DDA0008B

PowerPAD™ SOIC - 1.7 mm max height

PLASTIC SMALL OUTLINE



STENCIL THICKNESS	SOLDER STENCIL OPENING
0.1	3.03 X 3.80
0.125	2.71 X 3.40 (SHOWN)
0.150	2.47 X 3.10
0.175	2.29 X 2.87

4214849/B 09/2025

NOTES: (continued)

11. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
12. Board assembly site may have different recommendations for stencil design.

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